Step 1

```
# Import required libraries
import yfinance as yf
import pandas as pd
import numpy as np

def download_etf_data(etf_symbols, start_date, end_date):
    """
    Downloads daily adjusted closing prices for specified ETFs using yfinance.

    Parameters:
        etf_symbols (list): List of ETF ticker symbols to download
        start_date (str): Start date for data collection (format: 'YYYY-MM-DD')
        end_date (str): End date for data collection (format: 'YYYY-MM-DD')

    Returns:
        pd.DataFrame: DataFrame containing daily adjusted closing prices for each ETF
    """
    # Download data from yfinance (focus on adjusted close prices for accuracy)
    # Adjusted close accounts for dividends, stock splits, and other corporate actions
    data = yf.download(
        tickers=etf_symbols,
        start=start_date,
        end=end_date,
        interval='1d',  # Daily frequency (required by the task)
        progress=False,  # Disable download progress bar for cleaner output
        auto_adjust=False,  # We'll explicitly use 'Adj Close' column
        actions=False  # Actions (dividends/splits) are already reflected in Adj Close
    )

    # Extract only the Adjusted Close prices (most relevant for return calculations)
    adj_close_prices = data['Adj Close'].dropna()  # Remove rows with missing values

    # Print summary information for verification
    print(f"Data download completed successfully!")
```

```
    print(f"Date range: {adj_close_prices.index.min().strftime('%Y-%m-%d')} to {adj_close_prices.index.max().s
    print(f"Number of trading days: {len(adj_close_prices)}")
    print(f"ETFs included: {', '.join(adj_close_prices.columns.tolist())}")

    return adj_close_prices

if __name__ == "__main__":
    # Step 1: Define core parameters as per the task requirements
    # 5 ETFs representing different asset classes
    ETF_SYMBOLS = [
        "SPY",   # Equity (SPDR S&P500 ETF)
        "TLT",   # Fixed Income (iShares 20+ Year Treasury Bond ETF)
        "SHY",   # Cash-like (iShares 1-3 Year Treasury Bond ETF)
        "GLD",   # Precious metals (SPDR Gold Shares)
        "DBO"    # Crude Oil (Invesco DB Oil Fund)
    ]

    # Test sample period: January 1st, 2018 to December 30th, 2022
    TEST_START = "2018-01-01"
    TEST_END = "2022-12-30"

    # Step 2: Download test sample data
    test_sample_data = download_etf_data(ETF_SYMBOLS, TEST_START, TEST_END)

    # Step 3: Define and download training sample (discretionary split: 5 years before test period)
    # Rationale: Using 2013-2017 as training data provides sufficient history without overlap
    TRAIN_START = "2013-01-01"
    TRAIN_END = "2017-12-31"
    training_sample_data = download_etf_data(ETF_SYMBOLS, TRAIN_START, TRAIN_END)

    # Step 4: Optional - Define and download validation sample (6 months before test period)
    # Purpose: For hyperparameter tuning and avoiding overfitting
    VALID_START = "2017-07-01"
    VALID_END = "2017-12-31"
    validation_sample_data = download_etf_data(ETF_SYMBOLS, VALID_START, VALID_END)

    # Step 5: Save data to CSV for future use (optional but recommended)
    test_sample_data.to_csv("etf_test_sample_2018_2022.csv")
    training_sample_data.to_csv("etf_training_sample_2013_2017.csv")
```

```
validation_sample_data.to_csv("etf_validation_sample_2017.csv")

# Optional: Display first 5 rows of test sample to verify data structure
print("\nSample of test data (first 5 rows):")
print(test_sample_data.head())
```

```
Data download completed successfully!
Date range: 2018-01-02 to 2022-12-29
Number of trading days: 1258
ETFs included: DBO, GLD, SHY, SPY, TLT
Data download completed successfully!
Date range: 2013-01-02 to 2017-12-29
Number of trading days: 1259
ETFs included: DBO, GLD, SHY, SPY, TLT
Data download completed successfully!
Date range: 2017-07-03 to 2017-12-29
Number of trading days: 126
ETFs included: DBO, GLD, SHY, SPY, TLT

Sample of test data (first 5 rows):
Ticker           DBO         GLD        SHY         SPY         TLT
Date
2018-01-02  8.944057  125.150002  70.879150  237.909149  100.963882
2018-01-03  9.137156  124.820000  70.879150  239.413940  101.446671
2018-01-04  9.154711  125.459999  70.845352  240.423035  101.430573
2018-01-05  9.110826  125.330002  70.845352  242.025269  101.140915
2018-01-08  9.163488  125.309998  70.845352  242.467819  101.076553
```

```
# Import required libraries (extended for EDA/visualization)
import matplotlib.pyplot as plt
from statsmodels.tsa.stattools import adfuller
import seaborn as sns
from scipy import stats

# Set plot style and parameters for readability
plt.style.use('seaborn-v0_8-whitegrid')
plt.rcParams['figure.figsize'] = (15, 10)
plt.rcParams['font.size'] = 10
plt.rcParams['axes.grid'] = True
```

```
def download_etf_data(etf_symbols, start_date, end_date):
    """
    Downloads daily adjusted closing prices for specified ETFs using yfinance.

    Parameters:
        etf_symbols (list): List of ETF ticker symbols to download
        start_date (str): Start date for data collection (format: 'YYYY-MM-DD')
        end_date (str): End date for data collection (format: 'YYYY-MM-DD')

    Returns:
        pd.DataFrame: DataFrame containing daily adjusted closing prices for each ETF
    """
    data = yf.download(
        tickers=etf_symbols,
        start=start_date,
        end=end_date,
        interval='1d',
        progress=False,
        auto_adjust=False,
        actions=False
    )
    adj_close_prices = data['Adj Close'].dropna()

    # Print verification summary
    print(f"Data range: {adj_close_prices.index.min().strftime('%Y-%m-%d')} to {adj_close_prices.index.max().s
    print(f"Trading days: {len(adj_close_prices)}, ETFs: {', '.join(adj_close_prices.columns)}")

    return adj_close_prices

def calculate_daily_returns(price_data):
    """Calculates daily logarithmic returns (more suitable for time-series analysis than simple returns)"""
    return np.log(price_data / price_data.shift(1)).dropna()

def summary_statistics(returns_data):
    """Generates key summary statistics for daily returns"""
    stats_df = pd.DataFrame({
        'Mean Return (%)': returns_data.mean() * 100,
        'Volatility (%)': returns_data.std() * 100,
        'Sharpe Ratio (rfr=0%)': returns_data.mean() / returns_data.std() * np.sqrt(252),  # Annualized
```

```
            'Skewness': returns_data.skew(),
            'Kurtosis': returns_data.kurtosis(),
            'Jarque-Bera p-value': [stats.jarque_bera(returns_data[etf])[1] for etf in returns_data.columns]
        })
        print("\n=== Summary Statistics (Daily Returns) ===")
        print(stats_df.round(4))
        return stats_df

    def adf_stationarity_test(series, name):
        """Performs Augmented Dickey-Fuller test for stationarity"""
        result = adfuller(series)
        print(f"\nADF Test for {name}:")
        print(f"Test Statistic: {result[0]:.4f}")
        print(f"p-value: {result[1]:.4f}")
        print(f"Critical Values (1%/5%/10%): {result[4]}")
        print(f"Stationary: {'YES' if result[1] < 0.05 else 'NO'} (p<0.05)")

    def plot_price_series(price_data):
        """Plots normalized price time-series (to compare trends on same scale)"""
        # Normalize prices to 100 at start for better visual comparison
        normalized_prices = (price_data / price_data.iloc[0]) * 100

        plt.subplot(2, 2, 1)
        for etf in normalized_prices.columns:
            plt.plot(normalized_prices.index, normalized_prices[etf], label=etf, linewidth=1.5)
        plt.title('Normalized Adjusted Prices (2018-2022)', fontweight='bold')
        plt.xlabel('Date')
        plt.ylabel('Normalized Price (Start=100)')
        plt.legend()
        plt.tight_layout()

    def plot_returns_distribution(returns_data):
        """Plots histogram + KDE of daily returns (individual distribution)"""
        plt.subplot(2, 2, 2)
        for etf in returns_data.columns:
            sns.kdeplot(returns_data[etf], label=etf, linewidth=2)
        plt.title('Distribution of Daily Returns', fontweight='bold')
        plt.xlabel('Daily Return (%)')
        plt.ylabel('Density')
```

```
    plt.legend()
    plt.tight_layout()

def plot_correlation_heatmap(returns_data):
    """Plots correlation matrix of daily returns (joint distribution)"""
    corr_matrix = returns_data.corr()

    plt.subplot(2, 2, 3)
    sns.heatmap(
        corr_matrix,
        annot=True,
        cmap='coolwarm',
        vmin=-1,
        vmax=1,
        fmt='.2f',
        square=True,
        cbar_kws={'shrink': 0.8}
    )
    plt.title('Correlation Matrix (Daily Returns)', fontweight='bold')
    plt.tight_layout()

def plot_volatility_clustering(returns_data):
    """Plots rolling volatility to show clustering (time-series characteristic)"""
    # 20-day rolling window (approx 1 month of trading days)
    rolling_vol = returns_data.rolling(window=20).std() * np.sqrt(252) * 100  # Annualized

    plt.subplot(2, 2, 4)
    for etf in rolling_vol.columns:
        plt.plot(rolling_vol.index, rolling_vol[etf], label=etf, linewidth=1.5)
    plt.title('Rolling Volatility (20-day Window, Annualized)', fontweight='bold')
    plt.xlabel('Date')
    plt.ylabel('Volatility (%)')
    plt.legend()
    plt.tight_layout()

if __name__ == "__main__":
    # ----------------------
    # Step 1: Download Data (Reused from previous script)
    # ----------------------
```

```
ETF_SYMBOLS = ["SPY", "TLT", "SHY", "GLD", "DBO"]
TEST_START = "2018-01-01"
TEST_END = "2022-12-30"

# Download test sample data (focus EDA on test period as per task)
price_data = download_etf_data(ETF_SYMBOLS, TEST_START, TEST_END)

# Calculate daily returns (core for EDA beyond price trends)
returns_data = calculate_daily_returns(price_data)

# ----------------------
# Step 2: Core EDA
# ----------------------
# 1. Summary Statistics
summary_stats = summary_statistics(returns_data)

# 2. Stationarity Test (ADF Test)
print("\n=== Stationarity Test (Augmented Dickey-Fuller) ===")
for etf in returns_data.columns:
    adf_stationarity_test(returns_data[etf], etf)

# 3. Visualizations (4 key plots in one figure)
plt.figure(figsize=(16, 12))

plot_price_series(price_data)          # Trend comparison (normalized)
plot_returns_distribution(returns_data)# Individual return distributions
plot_correlation_heatmap(returns_data) # Joint distribution (correlations)
plot_volatility_clustering(returns_data)# Time-series volatility clustering

plt.suptitle('ETF Exploratory Data Analysis (2018-2022)', fontsize=16, fontweight='bold')
plt.tight_layout(rect=[0, 0, 1, 0.96])  # Adjust for suptitle
plt.savefig('etf_eda_2018_2022.png', dpi=300, bbox_inches='tight')
plt.show()

# ----------------------
# Optional: Seasonality Check (Monthly Returns)
# ----------------------
returns_data['Month'] = returns_data.index.month
monthly_returns = returns_data.groupby('Month').mean() * 100
```

```
plt.figure(figsize=(12, 6))
monthly_returns.plot(kind='bar', ax=plt.gca())
plt.title('Average Monthly Returns (%) - Seasonality Check', fontweight='bold')
plt.xlabel('Month (1=Jan, 12=Dec)')
plt.ylabel('Average Daily Return (%)')
plt.xticks(rotation=0)
plt.legend(title='ETF')
plt.grid(axis='y', alpha=0.3)
plt.savefig('etf_monthly_seasonality.png', dpi=300, bbox_inches='tight')
plt.show()
```

```
Data range: 2018-01-02 to 2022-12-29
Trading days: 1258, ETFs: DBO, GLD, SHY, SPY, TLT

=== Summary Statistics (Daily Returns) ===
        Mean Return (%)  Volatility (%)  Sharpe Ratio (rfr=0%)  Skewness  \
Ticker
DBO              0.0332          2.1855                 0.2415   -1.1256
GLD              0.0238          0.9057                 0.4176   -0.3445
SHY              0.0026          0.0882                 0.4652    0.0837
SPY              0.0351          1.3638                 0.4081   -0.7804
TLT             -0.0093          1.0162                -0.1453    0.1189

        Kurtosis  Jarque-Bera p-value
Ticker
DBO       7.2558                  0.0
GLD       3.6419                  0.0
SHY       7.5154                  0.0
SPY      11.2261                  0.0
```

Step 2

```
# Import required libraries
import yfinance as yf
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.tsa.stattools import adfuller
from scipy import stats
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, r2_score, mean_absolute_error
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping
import warnings
warnings.filterwarnings('ignore')

# Set random seeds for reproducibility
np.random.seed(42)
tf.random.set_seed(42)
```

```
# Configure plot style for readability
plt.style.use('seaborn-v0_8-whitegrid')
plt.rcParams['figure.figsize'] = (15, 10)
plt.rcParams['font.size'] = 10

# ----------------------
# Reusable Data Loading Functions
# ----------------------
def download_etf_data(etf_symbols, start_date, end_date):
    """
    Downloads daily adjusted closing prices for specified ETFs using yfinance.

    Parameters:
        etf_symbols (list): List of ETF ticker symbols
        start_date (str): Start date in 'YYYY-MM-DD' format
        end_date (str): End date in 'YYYY-MM-DD' format

    Returns:
        pd.DataFrame: DataFrame with adjusted closing prices (index: dates, columns: ETFs)
    """
    data = yf.download(
        tickers=etf_symbols,
        start=start_date,
        end=end_date,
        interval='1d',
        progress=False,
        auto_adjust=False,
        actions=False
    )
    adj_close_prices = data['Adj Close'].dropna()
    print(f"Data range: {adj_close_prices.index.min().strftime('%Y-%m-%d')} to {adj_close_prices.index.max().s
    print(f"Trading days: {len(adj_close_prices)}, ETFs: {', '.join(adj_close_prices.columns)}")
    return adj_close_prices

def calculate_daily_returns(price_data):
    """
    Calculates daily logarithmic returns (more suitable for time-series analysis than simple returns).
```

```
    Parameters:
        price_data (pd.DataFrame): Adjusted closing prices of ETFs

    Returns:
        pd.DataFrame: Daily log returns (index: dates, columns: ETFs)
    """
    return np.log(price_data / price_data.shift(1)).dropna()

# ----------------------
# Target Variable & Sequence Creation (FIXED LOGIC)
# ----------------------
def create_target_variable(returns_data, target_horizon=25):
    """
    Creates target variable: 25-day ahead cumulative log return for each ETF.
    Target = Sum of returns from t+1 to t+25 (cumulative return over next 25 trading days)

    Parameters:
        returns_data (pd.DataFrame): Daily log returns of ETFs
        target_horizon (int): Number of days ahead to predict cumulative return (default: 25)

    Returns:
        pd.DataFrame: 25-day ahead cumulative returns (index: dates, columns: ETFs)
    """
    targets = {}
    for etf in returns_data.columns:
        # Calculate rolling cumulative return for future 'target_horizon' days
        cumulative_future_return = returns_data[etf].rolling(window=target_horizon).sum().shift(-target_horizo
        targets[etf] = cumulative_future_return
    return pd.DataFrame(targets).dropna()

def create_lstm_sequences(returns_data, target_data, etf, lag_window=30):
    """
    Creates properly aligned input sequences (X) and target values (y) for LSTM training.
    FIXED: Ensures 1:1 mapping between input sequences and target values (no cardinality mismatch)

    Parameters:
        returns_data (pd.DataFrame): Daily log returns of all ETFs
        target_data (pd.DataFrame): 25-day ahead cumulative returns of all ETFs
        etf (str): Specific ETF ticker to create sequences for
```

```
        lag_window (int): Number of past days to use as input features (default: 30)

    Returns:
        tuple: (X, y, scaler)
            X: 3D array of input sequences (shape: [n_samples, lag_window, 1])
            y: 2D array of target values (shape: [n_samples, 1])
            scaler: StandardScaler fitted on the input features (for inference)
    """
    # Step 1: Align features (past returns) and target (25-day ahead return) by date
    # Use inner join to ensure only dates with both features and targets are kept
    features = returns_data[[etf]].rename(columns={etf: 'daily_return'})
    targets = target_data[[etf]].rename(columns={etf: 'cumulative_25d_return'})
    aligned_data = features.join(targets, how='inner').dropna()

    # Step 2: Validate data sufficiency
    if len(aligned_data) <= lag_window:
        raise ValueError(
            f"Insufficient data for {etf}: Aligned data length ({len(aligned_data)}) "
            f"must be greater than lag window ({lag_window})"
        )

    # Step 3: Extract raw features and targets (guaranteed to have same length)
    X_raw = aligned_data['daily_return'].values  # Shape: (n_aligned_samples,)
    y = aligned_data['cumulative_25d_return'].values  # Shape: (n_aligned_samples,)
    print(f"Aligned data for {etf}: {len(X_raw)} samples (features = targets)")

    # Step 4: Standardize features (critical for neural network stability)
    scaler = StandardScaler()
    X_scaled = scaler.fit_transform(X_raw.reshape(-1, 1))  # Reshape to 2D for scaler: (n_samples, 1)

    # Step 5: Create sliding window sequences (core fixed logic)
    X_sequences = []
    # Loop to create sequences: each sequence = [t-lag_window+1, ..., t] -> predicts target at t+1
    for i in range(lag_window - 1, len(X_scaled) - 1):
        # Extract window of past 'lag_window' days of scaled returns
        sequence = X_scaled[i - lag_window + 1 : i + 1, 0]  # Shape: (lag_window,)
        X_sequences.append(sequence)

    # Step 6: Convert to LSTM-compatible shapes
```

```
    X = np.array(X_sequences).reshape(-1, lag_window, 1)  # (n_samples, time_steps, features)
    y = y[lag_window:].reshape(-1, 1)  # Align targets to sequences (skip first 'lag_window' targets)

    # Critical validation: Ensure X and y have matching number of samples
    assert len(X) == len(y), f"Sequence-target mismatch: X={len(X)} samples, y={len(y)} samples"
    print(f"Sequence creation successful for {etf}: X={X.shape}, y={y.shape}")

    return X, y, scaler

# ----------------------
# LSTM Model Architecture & Training
# ----------------------
def build_lstm_model(input_shape):
    """
    Builds a robust LSTM model with dropout regularization for time-series regression.

    Parameters:
        input_shape (tuple): Shape of input sequences (lag_window, 1)

    Returns:
        tf.keras.Model: Compiled LSTM model
    """
    model = Sequential([
        # First LSTM layer: captures short-term temporal dependencies
        LSTM(units=128, return_sequences=True, input_shape=input_shape, kernel_initializer='he_uniform'),
        Dropout(0.15),  # Reduce overfitting by randomly dropping 20% of neurons

        # Second LSTM layer: captures long-term temporal dependencies
        LSTM(units=64, return_sequences=False, kernel_initializer='he_uniform'),
        Dropout(0.15),

        # Dense layers for non-linear transformation
        Dense(units=32, activation='relu', kernel_initializer='he_uniform'),
        Dense(units=1)  # Output layer: no activation (regression task)
    ])

    # Compile model with Adam optimizer and MSE loss (standard for regression)
    model.compile(
        optimizer=Adam(learning_rate=0.001, clipnorm=1.0),  # Clipnorm prevents exploding gradients
```

```
        loss='mean_squared_error',
        metrics=['mean_absolute_error']
    )

    return model

def train_etf_lstm(etf, returns_data, target_data, lag_window=30, epochs=30, batch_size=32):
    """
    End-to-end training pipeline for single ETF LSTM model (25-day ahead return prediction).

    Parameters:
        etf (str): ETF ticker to train model for
        returns_data (pd.DataFrame): Daily log returns of all ETFs
        target_data (pd.DataFrame): 25-day ahead cumulative returns of all ETFs
        lag_window (int): Number of past days to use as input features
        epochs (int): Maximum number of training epochs
        batch_size (int): Batch size for gradient descent

    Returns:
        tuple: (model, history, X_val, y_val, y_pred_val)
            model: Trained LSTM model
            history: Training history (loss/metrics over epochs)
            X_val: Validation input sequences
            y_val: Actual validation targets
            y_pred_val: Predicted validation targets
    """
    print(f"\n=== Training LSTM Model for {etf} ===")

    # Step 1: Create properly aligned sequences
    X, y, scaler = create_lstm_sequences(returns_data, target_data, etf, lag_window)

    # Step 2: Time-based train-validation split (no shuffle to preserve temporal order)
    train_size = int(0.8 * len(X))
    # Ensure minimum validation samples (avoid edge cases with small datasets)
    if train_size >= len(X) - 10:
        train_size = len(X) - 20  # Reserve at least 20 samples for validation
        print(f"Adjusted train size to {train_size} (small dataset safeguard)")

    X_train, X_val = X[:train_size], X[train_size:]
```

```
y_train, y_val = y[:train_size], y[train_size:]
print(f"Train set: X={X_train.shape}, y={y_train.shape}; Validation set: X={X_val.shape}, y={y_val.shape}"

# Step 3: Build model
input_shape = (X_train.shape[1], X_train.shape[2])  # (lag_window, 1)
model = build_lstm_model(input_shape)
model.summary()

# Step 4: Early stopping to prevent overfitting
early_stopping = EarlyStopping(
    monitor='val_loss',
    patience=8,  # Stop if validation loss doesn't improve for 8 epochs
    restore_best_weights=True,  # Revert to best model weights
    verbose=1
)

# Step 5: Train model
history = model.fit(
    X_train, y_train,
    validation_data=(X_val, y_val),
    epochs=epochs,
    batch_size=batch_size,
    callbacks=[early_stopping],
    verbose=1,
    shuffle=False  # No shuffling for time-series data
)

# Step 6: Evaluate model on validation set
y_pred_val = model.predict(X_val, verbose=0)
mse = mean_squared_error(y_val, y_pred_val)
rmse = np.sqrt(mse)
mae = mean_absolute_error(y_val, y_pred_val)
r2 = r2_score(y_val, y_pred_val)

print(f"\nValidation Metrics for {etf}:")
print(f"RMSE: {rmse:.6f} | MAE: {mae:.6f} | R² Score: {r2:.4f}")

# Step 7: Save trained model and scaler for future inference
model.save(f'lstm_model_{etf}.h5')
```

```
    np.save(f'scaler_{etf}.npy', scaler)

    return model, history, X_val, y_val, y_pred_val

# ----------------------
# Visualization Functions
# ----------------------
def plot_training_history(history, etf):
    """
    Plots training vs validation loss and MAE over epochs.

    Parameters:
        history (tf.keras.callbacks.History): Training history object
        etf (str): ETF ticker for plot labeling
    """
    plt.figure(figsize=(12, 4))

    # Plot Loss
    plt.subplot(1, 2, 1)
    plt.plot(history.history['loss'], label='Training Loss (MSE)', color='blue')
    plt.plot(history.history['val_loss'], label='Validation Loss (MSE)', color='orange')
    plt.title(f'{etf} - Training vs Validation Loss')
    plt.xlabel('Epoch')
    plt.ylabel('Mean Squared Error')
    plt.legend()
    plt.grid(True, alpha=0.3)

    # Plot MAE
    plt.subplot(1, 2, 2)
    plt.plot(history.history['mean_absolute_error'], label='Training MAE', color='blue')
    plt.plot(history.history['val_mean_absolute_error'], label='Validation MAE', color='orange')
    plt.title(f'{etf} - Training vs Validation MAE')
    plt.xlabel('Epoch')
    plt.ylabel('Mean Absolute Error')
    plt.legend()
    plt.grid(True, alpha=0.3)

    plt.tight_layout()
    plt.savefig(f'training_history_{etf}.png', dpi=300, bbox_inches='tight')
```

```
    plt.show()

def plot_prediction_results(y_val, y_pred_val, etf):
    """
    Plots scatter plot of actual vs predicted 25-day ahead returns.

    Parameters:
        y_val (np.array): Actual validation targets
        y_pred_val (np.array): Predicted validation targets
        etf (str): ETF ticker for plot labeling
    """
    plt.figure(figsize=(10, 6))
    plt.scatter(y_val, y_pred_val, alpha=0.6, color='navy', label='Actual vs Predicted')
    # Plot perfect prediction line (y=x)
    min_val = min(y_val.min(), y_pred_val.min())
    max_val = max(y_val.max(), y_pred_val.max())
    plt.plot([min_val, max_val], [min_val, max_val], 'r--', label='Perfect Prediction (y=x)')

    plt.title(f'{etf} - Actual vs Predicted 25-Day Ahead Cumulative Return')
    plt.xlabel('Actual 25-Day Cumulative Return')
    plt.ylabel('Predicted 25-Day Cumulative Return')
    plt.legend()
    plt.grid(True, alpha=0.3)
    plt.savefig(f'prediction_results_{etf}.png', dpi=300, bbox_inches='tight')
    plt.show()

def plot_performance_summary(all_metrics):
    """
    Plots bar charts comparing RMSE and R² across all ETF models.

    Parameters:
        all_metrics (list): List of dictionaries with 'ETF', 'RMSE', 'R² Score'
    """
    metrics_df = pd.DataFrame(all_metrics)

    plt.figure(figsize=(14, 6))

    # RMSE Comparison
    plt.subplot(1, 2, 1)
```

```
    sns.barplot(x='ETF', y='RMSE', data=metrics_df, palette='viridis')
    plt.title('RMSE Comparison Across ETFs (Lower = Better)')
    plt.ylabel('Root Mean Squared Error')
    plt.grid(axis='y', alpha=0.3)

    # R² Score Comparison
    plt.subplot(1, 2, 2)
    sns.barplot(x='ETF', y='R² Score', data=metrics_df, palette='viridis')
    plt.title('R² Score Comparison Across ETFs (Higher = Better)')
    plt.ylabel('R² Score (Explained Variance)')
    plt.grid(axis='y', alpha=0.3)

    plt.tight_layout()
    plt.savefig('model_performance_summary.png', dpi=300, bbox_inches='tight')
    plt.show()

# ----------------------
# Main Execution Pipeline
# ----------------------
if __name__ == "__main__":
    # Step 1: Configuration Parameters
    ETF_SYMBOLS = ["SPY", "TLT", "SHY", "GLD", "DBO"]  # 5 ETFs across asset classes
    TEST_START = "2018-01-01"                          # Start of test period
    TEST_END = "2022-12-30"                            # End of test period
    TARGET_HORIZON = 25                                # Predict 25-day ahead return
    LAG_WINDOW = 30                                    # Use past 90 days of returns as input
    EPOCHS = 30                                        # Max training epochs (reduced to prevent overfitting)
    BATCH_SIZE = 32                                    # Batch size for training

    # Step 2: Load and Preprocess Data
    print("=== Loading and Preprocessing Data ===")
    price_data = download_etf_data(ETF_SYMBOLS, TEST_START, TEST_END)
    returns_data = calculate_daily_returns(price_data)
    print(f"Daily returns data shape: {returns_data.shape}")

    # Create target variable (25-day ahead cumulative return)
    target_data = create_target_variable(returns_data, TARGET_HORIZON)
    print(f"Target data shape: {target_data.shape}")
    print(f"Target variable: {TARGET_HORIZON}-day ahead cumulative log return")
```

```
# Step 3: Train LSTM Models for Each ETF
print("\n=== Starting LSTM Model Training for All ETFs ===")
all_models = {}  # Store trained models
all_metrics = [] # Store performance metrics

for etf in ETF_SYMBOLS:
    try:
        model, history, X_val, y_val, y_pred_val = train_etf_lstm(
            etf=etf,
            returns_data=returns_data,
            target_data=target_data,
            lag_window=LAG_WINDOW,
            epochs=EPOCHS,
            batch_size=BATCH_SIZE
        )

        # Save model and metrics
        all_models[etf] = model
        all_metrics.append({
            'ETF': etf,
            'RMSE': np.sqrt(mean_squared_error(y_val, y_pred_val)),
            'MAE': mean_absolute_error(y_val, y_pred_val),
            'R² Score': r2_score(y_val, y_pred_val)
        })

        # Generate visualizations
        plot_training_history(history, etf)
        plot_prediction_results(y_val, y_pred_val, etf)

    except Exception as e:
        print(f"\nError training model for {etf}: {str(e)}")
        continue

# Step 4: Generate Performance Summary
if all_metrics:
    print("\n=== LSTM Model Performance Summary (All ETFs) ===")
    metrics_df = pd.DataFrame(all_metrics).round(4)
    print(metrics_df)
```

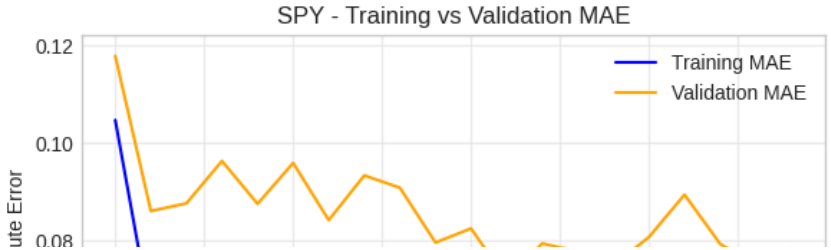
```
        plot_performance_summary(all_metrics)
    else:
        print("\nNo models were trained successfully.")
```

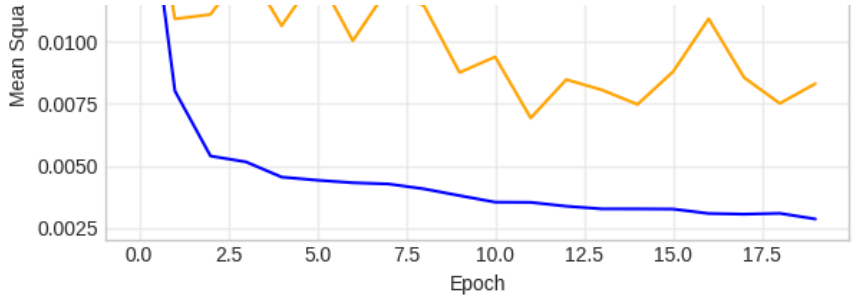
```
=== Loading and Preprocessing Data ===
Data range: 2018-01-02 to 2022-12-29
Trading days: 1258, ETFs: DBO, GLD, SHY, SPY, TLT
Daily returns data shape: (1257, 5)
Target data shape: (1232, 5)
Target variable: 25-day ahead cumulative log return

=== Starting LSTM Model Training for All ETFs ===

=== Training LSTM Model for SPY ===
Aligned data for SPY: 1232 samples (features = targets)
Sequence creation successful for SPY: X=(1202, 30, 1), y=(1202, 1)
Train set: X=(961, 30, 1), y=(961, 1); Validation set: X=(241, 30, 1), y=(241, 1)
Model: "sequential_37"
```

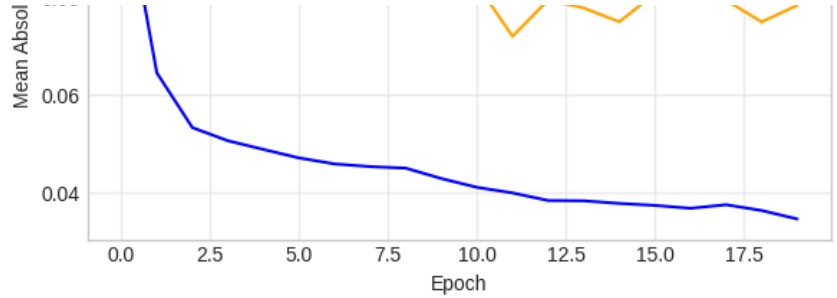
Layer (type)	Output Shape	Param #
lstm_74 (LSTM)	(None, 30, 128)	66,560
dropout_74 (Dropout)	(None, 30, 128)	0
lstm_75 (LSTM)	(None, 64)	49,408
dropout_75 (Dropout)	(None, 64)	0
dense_74 (Dense)	(None, 32)	2,080
dense_75 (Dense)	(None, 1)	33

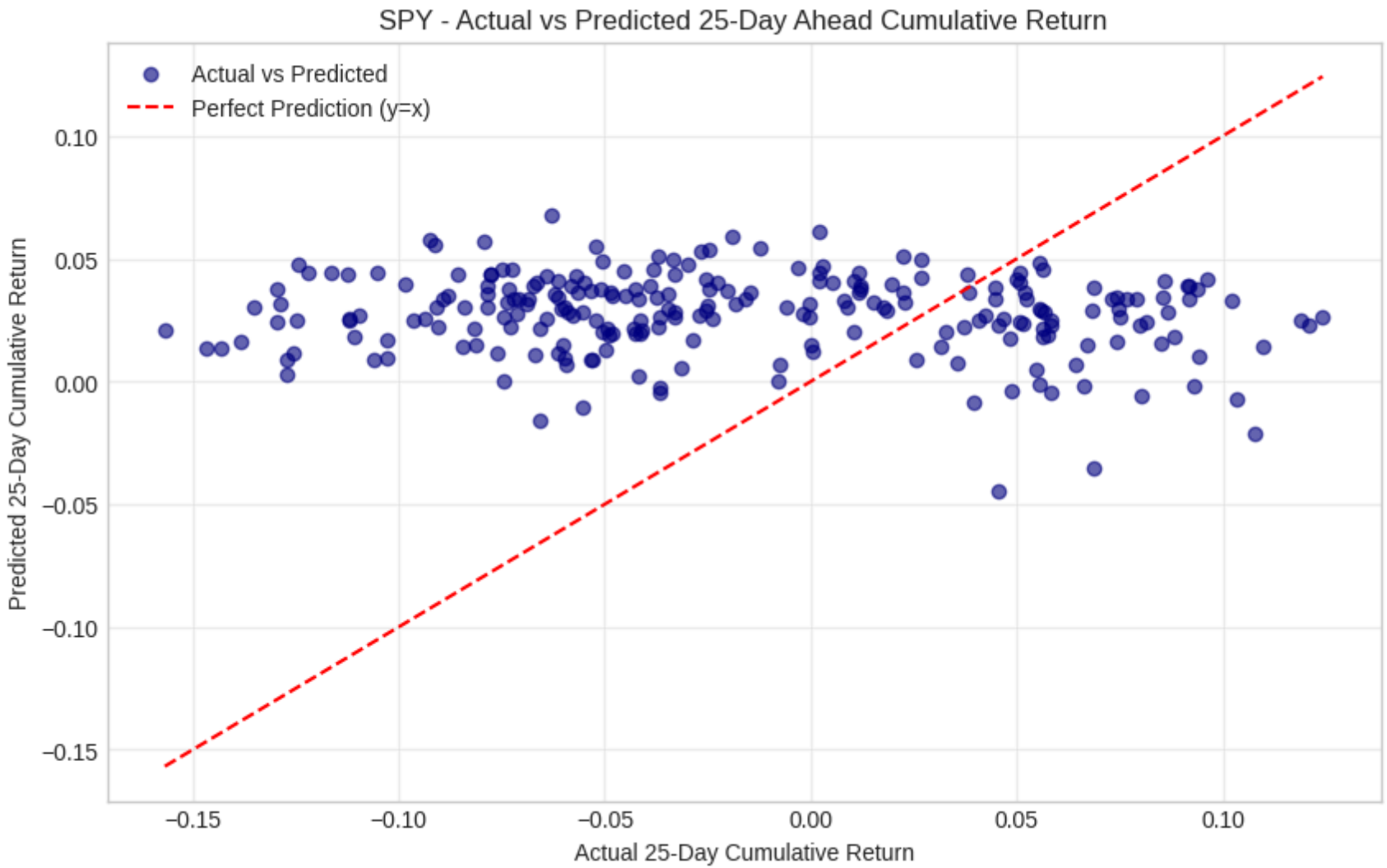
```
 Total params: 118,081 (461.25 KB)
 Trainable params: 118,081 (461.25 KB)
 Non-trainable params: 0 (0.00 B)
Epoch 1/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 9s 78ms/step - loss: 0.0269 - mean_absolute_error: 0.1235 - val_loss: 0.0192 - val_m
Epoch 2/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 63ms/step - loss: 0.0088 - mean_absolute_error: 0.0716 - val_loss: 0.0109 - val_m
Epoch 3/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 62ms/step - loss: 0.0058 - mean_absolute_error: 0.0574 - val_loss: 0.0111 - val_m
Epoch 4/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 63ms/step - loss: 0.0054 - mean_absolute_error: 0.0538 - val_loss: 0.0130 - val_m
Epoch 5/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 4s 99ms/step - loss: 0.0049 - mean_absolute_error: 0.0517 - val_loss: 0.0106 - val_m
Epoch 6/30
```

```
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 61ms/step - loss: 0.0043 - mean_absolute_error: 0.0481 - val_loss: 0.0127 - val_m
Epoch 7/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 63ms/step - loss: 0.0045 - mean_absolute_error: 0.0490 - val_loss: 0.0100 - val_m
Epoch 8/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 64ms/step - loss: 0.0041 - mean_absolute_error: 0.0462 - val_loss: 0.0122 - val_m
Epoch 9/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 64ms/step - loss: 0.0041 - mean_absolute_error: 0.0478 - val_loss: 0.0114 - val_m
Epoch 10/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 69ms/step - loss: 0.0040 - mean_absolute_error: 0.0461 - val_loss: 0.0088 - val_m
Epoch 11/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 99ms/step - loss: 0.0033 - mean_absolute_error: 0.0409 - val_loss: 0.0094 - val_m
Epoch 12/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 63ms/step - loss: 0.0033 - mean_absolute_error: 0.0397 - val_loss: 0.0069 - val_m
Epoch 13/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 62ms/step - loss: 0.0030 - mean_absolute_error: 0.0374 - val_loss: 0.0085 - val_m
Epoch 14/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 67ms/step - loss: 0.0030 - mean_absolute_error: 0.0385 - val_loss: 0.0081 - val_m
Epoch 15/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 63ms/step - loss: 0.0029 - mean_absolute_error: 0.0366 - val_loss: 0.0075 - val_m
Epoch 16/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 69ms/step - loss: 0.0029 - mean_absolute_error: 0.0360 - val_loss: 0.0088 - val_m
Epoch 17/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 96ms/step - loss: 0.0027 - mean_absolute_error: 0.0356 - val_loss: 0.0109 - val_m
Epoch 18/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 4s 62ms/step - loss: 0.0029 - mean_absolute_error: 0.0385 - val_loss: 0.0086 - val_m
Epoch 19/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 64ms/step - loss: 0.0028 - mean_absolute_error: 0.0357 - val_loss: 0.0075 - val_m
Epoch 20/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 69ms/step - loss: 0.0025 - mean_absolute_error: 0.0328 - val_loss: 0.0083 - val_m
Epoch 20: early stopping
Restoring model weights from the end of the best epoch: 12.
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`.

Validation Metrics for SPY:
RMSE: 0.083260 | MAE: 0.071991 | R² Score: -0.5809
```









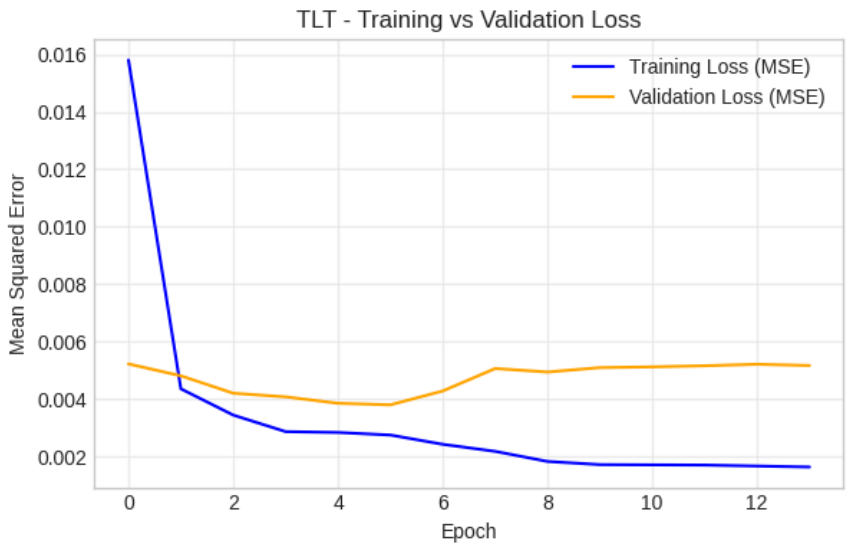

```
=== Training LSTM Model for TLT ===
Aligned data for TLT: 1232 samples (features = targets)
```

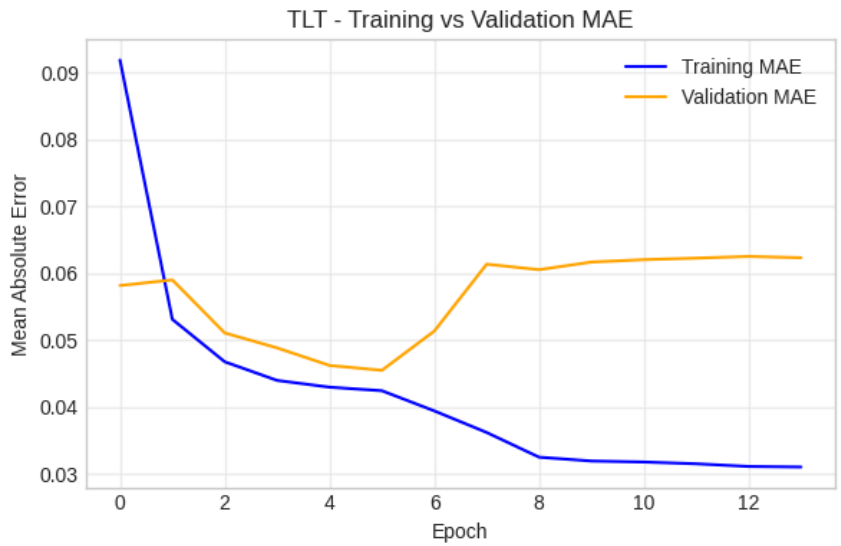
```
Sequence creation successful for TLT: X=(1202, 30, 1), y=(1202, 1)
Train set: X=(961, 30, 1), y=(961, 1); Validation set: X=(241, 30, 1), y=(241, 1)
Model: "sequential_38"
```

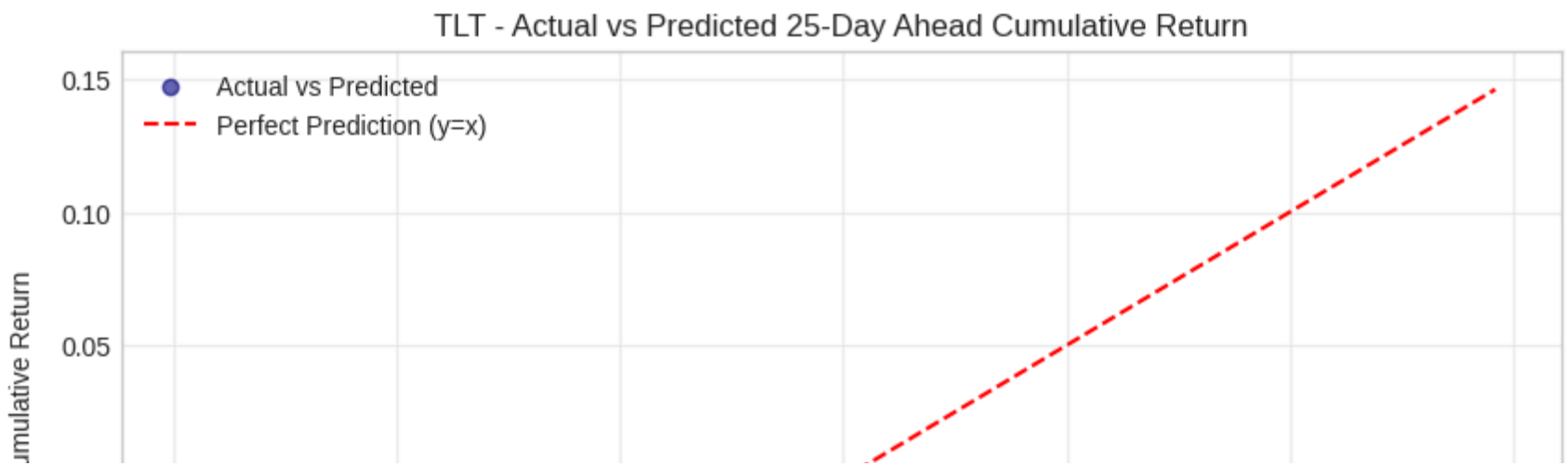
Layer (type)	Output Shape	Param #
lstm_76 (LSTM)	(None, 30, 128)	66,560
dropout_76 (Dropout)	(None, 30, 128)	0
lstm_77 (LSTM)	(None, 64)	49,408
dropout_77 (Dropout)	(None, 64)	0
dense_76 (Dense)	(None, 32)	2,080
dense_77 (Dense)	(None, 1)	33

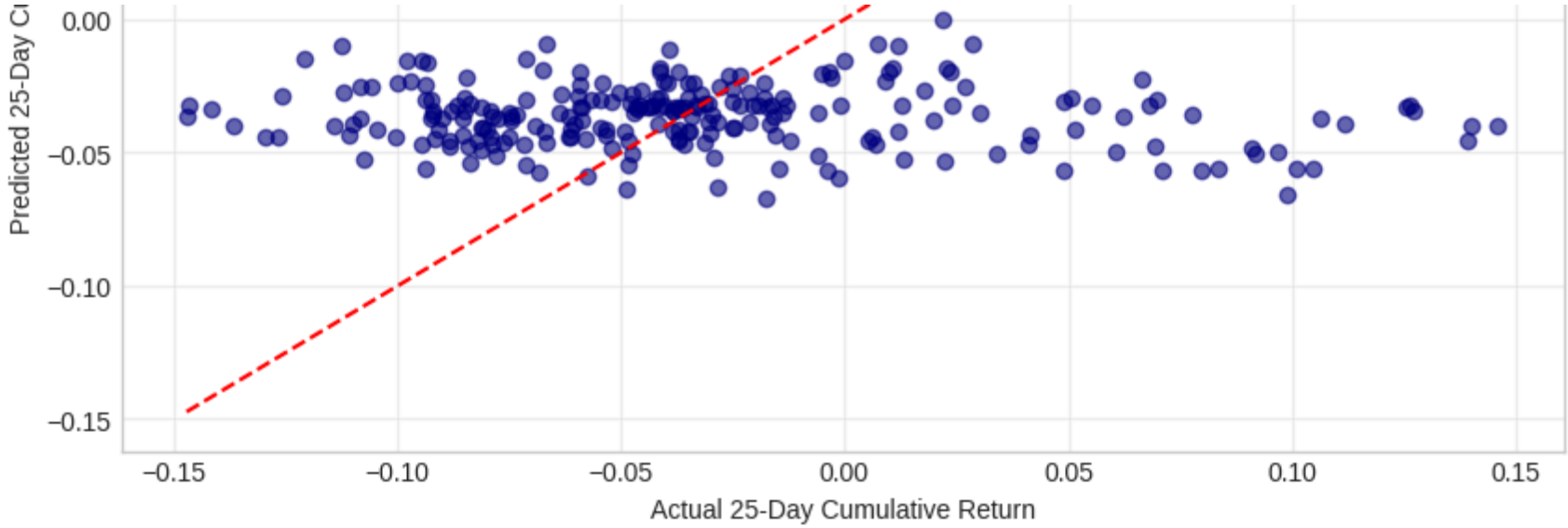
```
 Total params: 118,081 (461.25 KB)
 Trainable params: 118,081 (461.25 KB)
 Non-trainable params: 0 (0.00 B)
Epoch 1/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 6s 79ms/step - loss: 0.0266 - mean_absolute_error: 0.1191 - val_loss: 0.0052 - val_m
Epoch 2/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 65ms/step - loss: 0.0040 - mean_absolute_error: 0.0505 - val_loss: 0.0048 - val_m
Epoch 3/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 73ms/step - loss: 0.0030 - mean_absolute_error: 0.0432 - val_loss: 0.0042 - val_m
Epoch 4/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 100ms/step - loss: 0.0024 - mean_absolute_error: 0.0397 - val_loss: 0.0041 - val_
Epoch 5/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 4s 69ms/step - loss: 0.0023 - mean_absolute_error: 0.0377 - val_loss: 0.0039 - val_m
Epoch 6/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 64ms/step - loss: 0.0022 - mean_absolute_error: 0.0372 - val_loss: 0.0038 - val_m
Epoch 7/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 64ms/step - loss: 0.0021 - mean_absolute_error: 0.0359 - val_loss: 0.0043 - val_m
Epoch 8/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 98ms/step - loss: 0.0020 - mean_absolute_error: 0.0347 - val_loss: 0.0051 - val_m
Epoch 9/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 4s 65ms/step - loss: 0.0015 - mean_absolute_error: 0.0290 - val_loss: 0.0050 - val_m
Epoch 10/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 64ms/step - loss: 0.0014 - mean_absolute_error: 0.0288 - val_loss: 0.0051 - val_m
Epoch 11/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 65ms/step - loss: 0.0014 - mean_absolute_error: 0.0284 - val_loss: 0.0051 - val_m
```

```
Epoch 12/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 68ms/step - loss: 0.0014 - mean_absolute_error: 0.0278 - val_loss: 0.0052 - val_m
Epoch 13/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 4s 101ms/step - loss: 0.0014 - mean_absolute_error: 0.0277 - val_loss: 0.0052 - val_
Epoch 14/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 4s 64ms/step - loss: 0.0013 - mean_absolute_error: 0.0271 - val_loss: 0.0052 - val_m
Epoch 14: early stopping
Restoring model weights from the end of the best epoch: 6.
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`.

Validation Metrics for TLT:
RMSE: 0.061720 | MAE: 0.045475 | R² Score: -0.0880
```









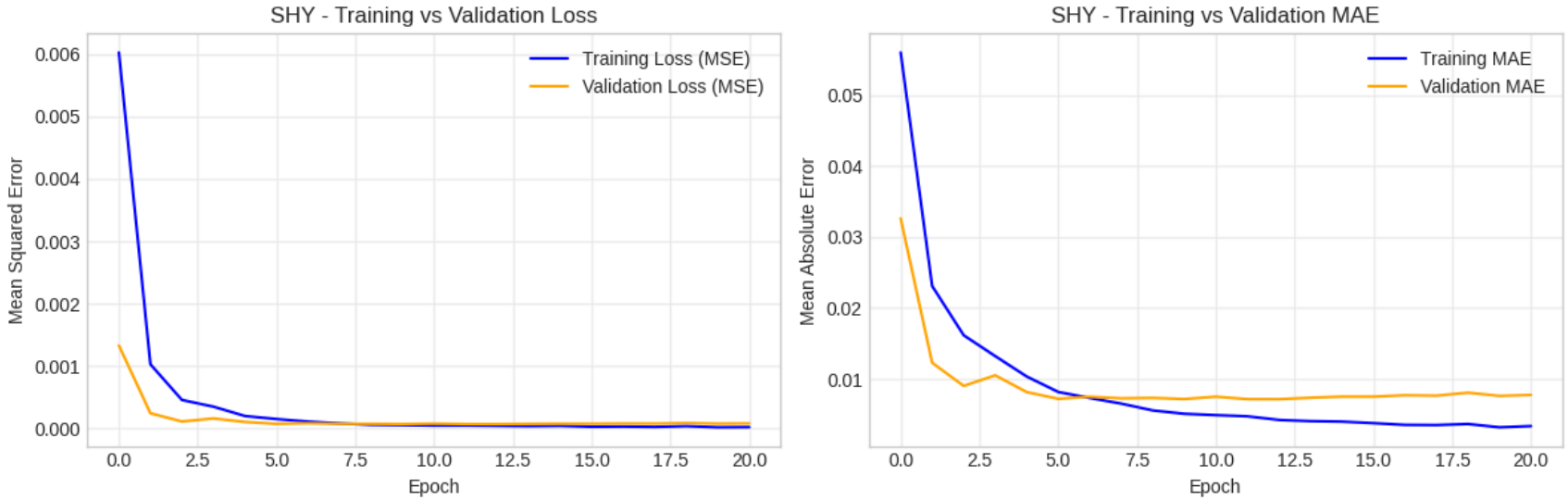

```
=== Training LSTM Model for SHY ===
Aligned data for SHY: 1232 samples (features = targets)
Sequence creation successful for SHY: X=(1202, 30, 1), y=(1202, 1)
Train set: X=(961, 30, 1), y=(961, 1); Validation set: X=(241, 30, 1), y=(241, 1)
Model: "sequential_39"
```

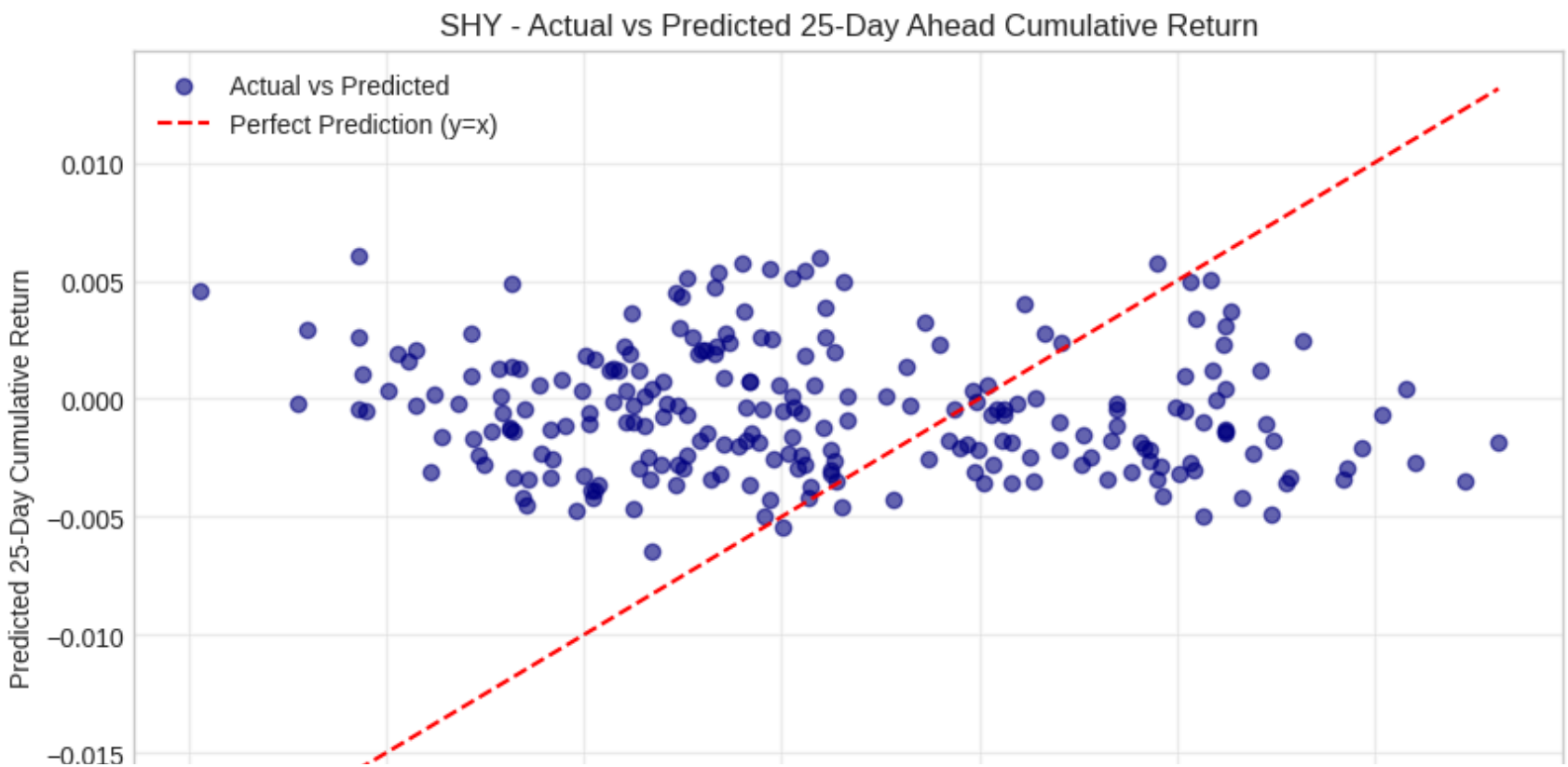
Layer (type)	Output Shape	Param #
lstm_78 (LSTM)	(None, 30, 128)	66,560
dropout_78 (Dropout)	(None, 30, 128)	0
lstm_79 (LSTM)	(None, 64)	49,408
dropout_79 (Dropout)	(None, 64)	0
dense_78 (Dense)	(None, 32)	2,080
dense_79 (Dense)	(None, 1)	33

```
 Total params: 118,081 (461.25 KB)
 Trainable params: 118,081 (461.25 KB)
 Non-trainable params: 0 (0.00 B)
Epoch 1/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 7s 87ms/step - loss: 0.0109 - mean_absolute_error: 0.0807 - val_loss: 0.0013 - val_m
Epoch 2/30
```

```
Epoch 2/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 66ms/step - loss: 0.0014 - mean_absolute_error: 0.0277 - val_loss: 2.4584e-04 - v
Epoch 3/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 67ms/step - loss: 5.6310e-04 - mean_absolute_error: 0.0185 - val_loss: 1.1428e-04
Epoch 4/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 66ms/step - loss: 3.9743e-04 - mean_absolute_error: 0.0147 - val_loss: 1.6255e-04
Epoch 5/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 87ms/step - loss: 2.1739e-04 - mean_absolute_error: 0.0110 - val_loss: 1.0638e-04
Epoch 6/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 88ms/step - loss: 1.6615e-04 - mean_absolute_error: 0.0085 - val_loss: 7.6776e-05
Epoch 7/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 66ms/step - loss: 1.1373e-04 - mean_absolute_error: 0.0076 - val_loss: 8.6089e-05
Epoch 8/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 67ms/step - loss: 8.9047e-05 - mean_absolute_error: 0.0069 - val_loss: 7.6095e-05
Epoch 9/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 66ms/step - loss: 6.5824e-05 - mean_absolute_error: 0.0060 - val_loss: 7.6461e-05
Epoch 10/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 66ms/step - loss: 5.7743e-05 - mean_absolute_error: 0.0053 - val_loss: 7.2841e-05
Epoch 11/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 104ms/step - loss: 4.9553e-05 - mean_absolute_error: 0.0050 - val_loss: 7.9285e-0
Epoch 12/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 70ms/step - loss: 4.6327e-05 - mean_absolute_error: 0.0048 - val_loss: 7.2764e-05
Epoch 13/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 65ms/step - loss: 5.6460e-05 - mean_absolute_error: 0.0047 - val_loss: 7.2193e-05
Epoch 14/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 66ms/step - loss: 3.9719e-05 - mean_absolute_error: 0.0042 - val_loss: 7.5937e-05
Epoch 15/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 67ms/step - loss: 4.3239e-05 - mean_absolute_error: 0.0042 - val_loss: 7.8976e-05
Epoch 16/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 80ms/step - loss: 3.0931e-05 - mean_absolute_error: 0.0039 - val_loss: 7.8688e-05
Epoch 17/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 101ms/step - loss: 4.6057e-05 - mean_absolute_error: 0.0038 - val_loss: 8.2452e-0
Epoch 18/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 66ms/step - loss: 2.6034e-05 - mean_absolute_error: 0.0036 - val_loss: 8.0867e-05
Epoch 19/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 67ms/step - loss: 3.3267e-05 - mean_absolute_error: 0.0036 - val_loss: 8.9803e-05
Epoch 20/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 66ms/step - loss: 2.0178e-05 - mean_absolute_error: 0.0033 - val_loss: 8.0357e-05
Epoch 21/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 65ms/step - loss: 2.4457e-05 - mean_absolute_error: 0.0034 - val_loss: 8.3418e-05
Epoch 21: early stopping
Restoring model weights from the end of the best epoch: 13.
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`.
```

```
Validation Metrics for SHY:
RMSE: 0.008497 | MAE: 0.007179 | R² Score: -0.5149
```



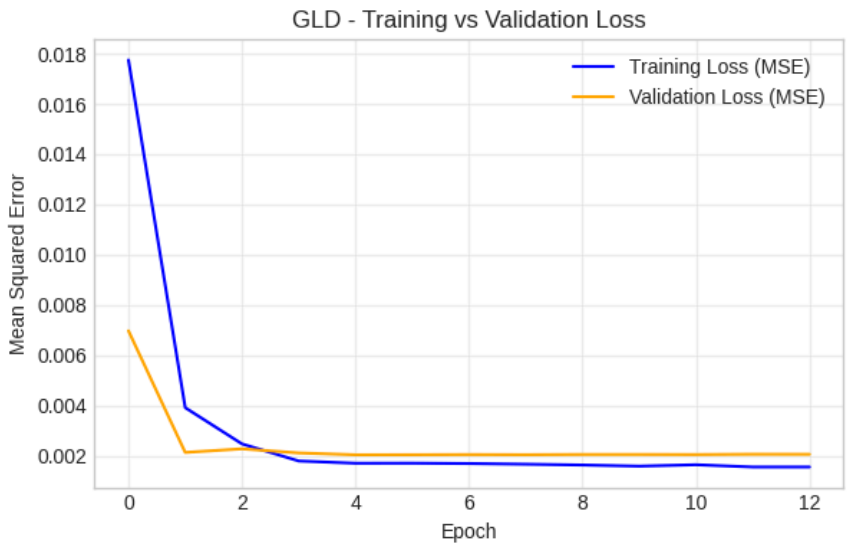


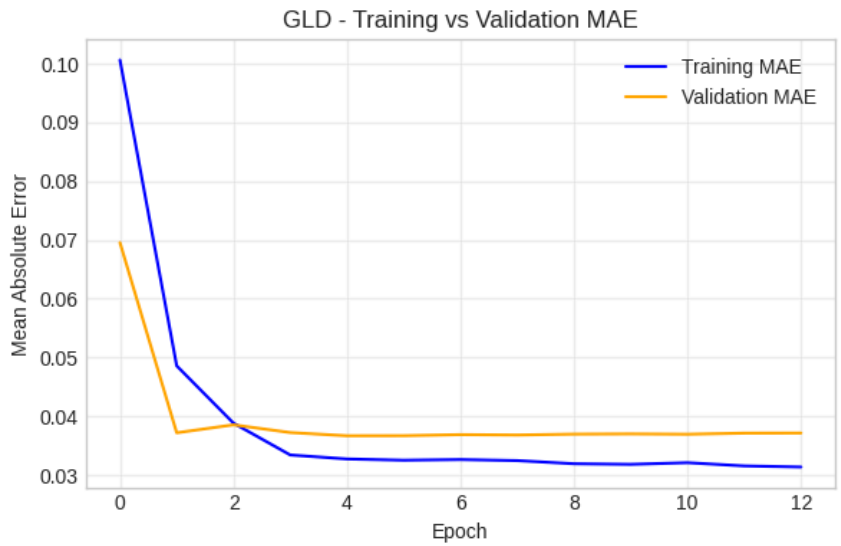
```
=== Training LSTM Model for GLD ===
Aligned data for GLD: 1232 samples (features = targets)
Sequence creation successful for GLD: X=(1202, 30, 1), y=(1202, 1)
Train set: X=(961, 30, 1), y=(961, 1); Validation set: X=(241, 30, 1), y=(241, 1)
Model: "sequential_40"
```

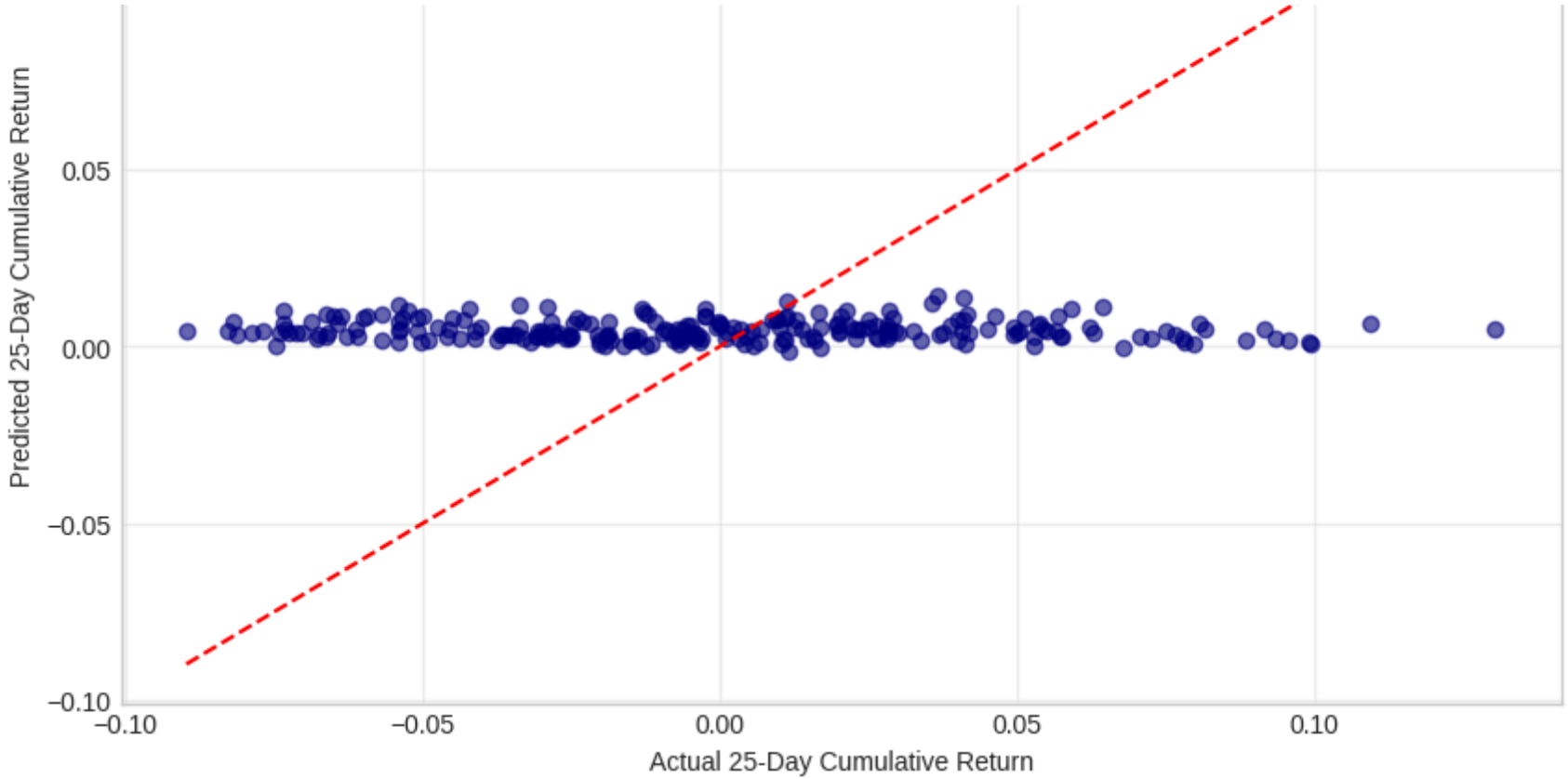
Layer (type)	Output Shape	Param #
lstm_80 (LSTM)	(None, 30, 128)	66,560
dropout_80 (Dropout)	(None, 30, 128)	0
lstm_81 (LSTM)	(None, 64)	49,408
dropout_81 (Dropout)	(None, 64)	0
dense_80 (Dense)	(None, 32)	2,080
dense_81 (Dense)	(None, 1)	33

```
 Total params: 118,081 (461.25 KB)
 Trainable params: 118,081 (461.25 KB)
 Non-trainable params: 0 (0.00 B)
Epoch 1/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 9s 85ms/step - loss: 0.0325 - mean_absolute_error: 0.1366 - val_loss: 0.0070 - val_m
Epoch 2/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 83ms/step - loss: 0.0044 - mean_absolute_error: 0.0518 - val_loss: 0.0021 - val_m
Epoch 3/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 98ms/step - loss: 0.0022 - mean_absolute_error: 0.0362 - val_loss: 0.0023 - val_m
Epoch 4/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 4s 69ms/step - loss: 0.0015 - mean_absolute_error: 0.0301 - val_loss: 0.0021 - val_m
Epoch 5/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 73ms/step - loss: 0.0014 - mean_absolute_error: 0.0296 - val_loss: 0.0020 - val_m
Epoch 6/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 69ms/step - loss: 0.0014 - mean_absolute_error: 0.0292 - val_loss: 0.0020 - val_m
```

```
Epoch 7/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 107ms/step - loss: 0.0014 - mean_absolute_error: 0.0295 - val_loss: 0.0020 - val_
Epoch 8/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 4s 68ms/step - loss: 0.0013 - mean_absolute_error: 0.0289 - val_loss: 0.0020 - val_m
Epoch 9/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 68ms/step - loss: 0.0013 - mean_absolute_error: 0.0285 - val_loss: 0.0021 - val_m
Epoch 10/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 69ms/step - loss: 0.0013 - mean_absolute_error: 0.0282 - val_loss: 0.0021 - val_m
Epoch 11/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 70ms/step - loss: 0.0013 - mean_absolute_error: 0.0283 - val_loss: 0.0020 - val_m
Epoch 12/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 4s 101ms/step - loss: 0.0012 - mean_absolute_error: 0.0280 - val_loss: 0.0021 - val_
Epoch 13/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 69ms/step - loss: 0.0012 - mean_absolute_error: 0.0277 - val_loss: 0.0021 - val_m
Epoch 13: early stopping
Restoring model weights from the end of the best epoch: 5.
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`.

Validation Metrics for GLD:
RMSE: 0.045170 | MAE: 0.036677 | R² Score: -0.0273
```





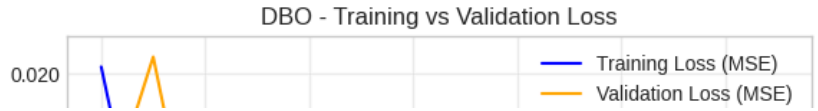


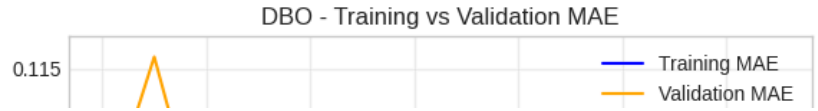

```
=== Training LSTM Model for DBO ===
Aligned data for DBO: 1232 samples (features = targets)
Sequence creation successful for DBO: X=(1202, 30, 1), y=(1202, 1)
Train set: X=(961, 30, 1), y=(961, 1); Validation set: X=(241, 30, 1), y=(241, 1)
```

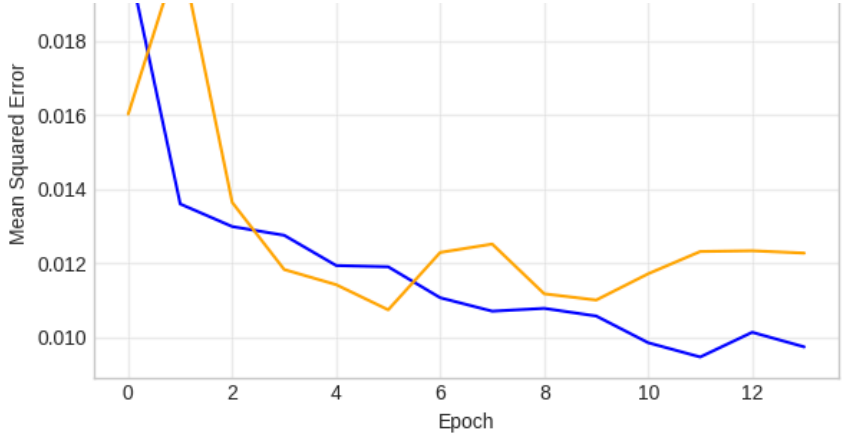
Model: "sequential_41"

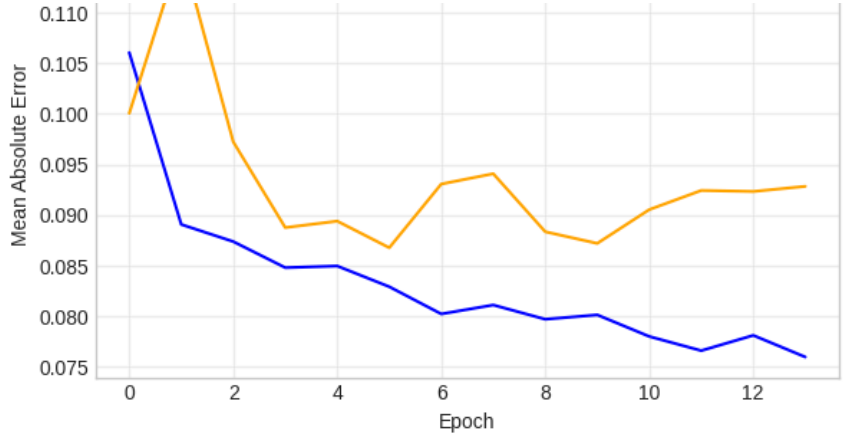
Layer (type)	Output Shape	Param #
lstm_82 (LSTM)	(None, 30, 128)	66,560
dropout_82 (Dropout)	(None, 30, 128)	0
lstm_83 (LSTM)	(None, 64)	49,408
dropout_83 (Dropout)	(None, 64)	0
dense_82 (Dense)	(None, 32)	2,080

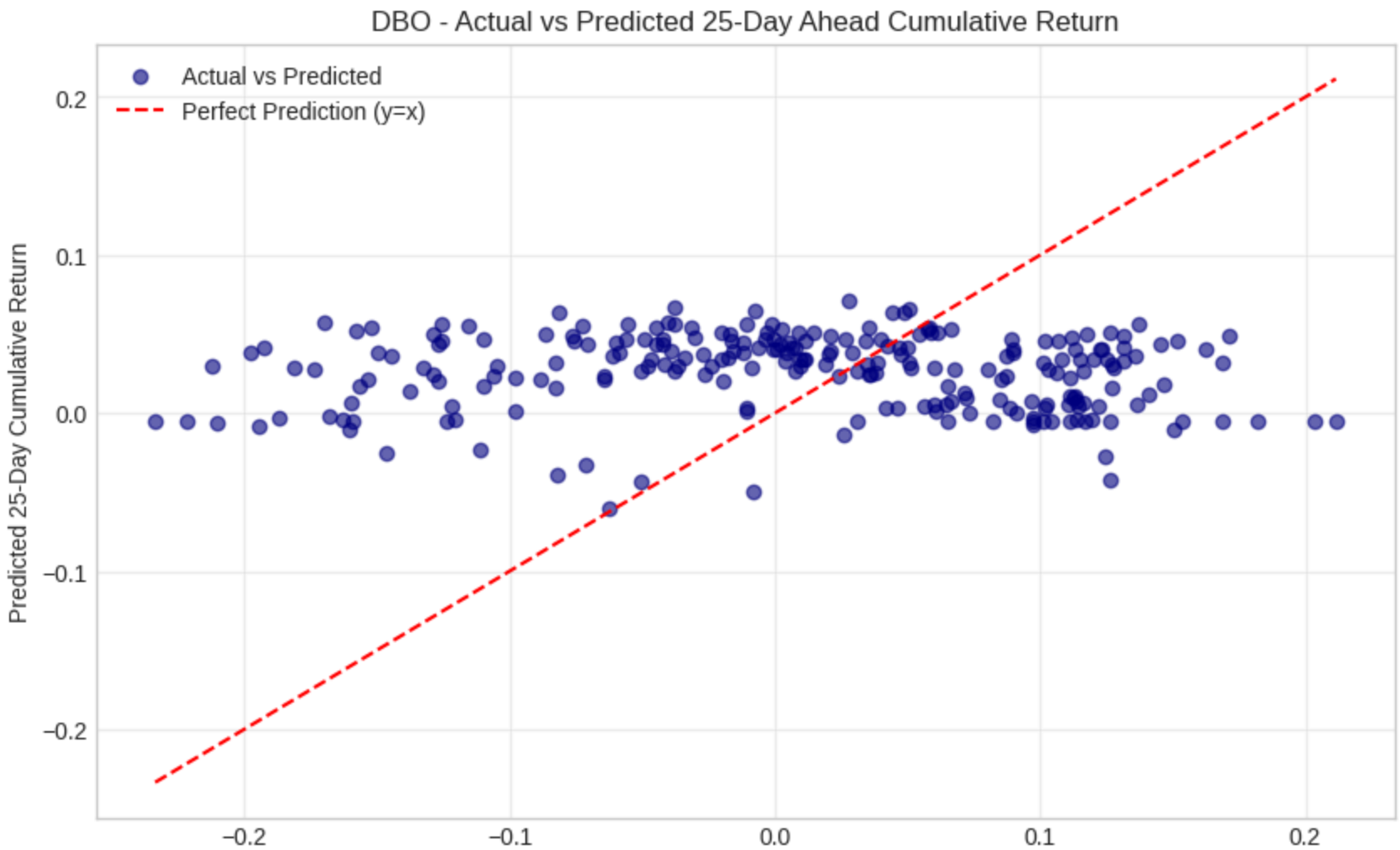
```
│ dense_83 (Dense)                │ (None, 1)               │            33 │
└─────────────────────────────────┴─────────────────────────┴───────────────┘

 Total params: 118,081 (461.25 KB)
 Trainable params: 118,081 (461.25 KB)
 Non-trainable params: 0 (0.00 B)
Epoch 1/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 8s 127ms/step - loss: 0.0210 - mean_absolute_error: 0.1052 - val_loss: 0.0160 - val_
Epoch 2/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 70ms/step - loss: 0.0119 - mean_absolute_error: 0.0818 - val_loss: 0.0205 - val_m
Epoch 3/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 70ms/step - loss: 0.0111 - mean_absolute_error: 0.0770 - val_loss: 0.0136 - val_m
Epoch 4/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 69ms/step - loss: 0.0112 - mean_absolute_error: 0.0752 - val_loss: 0.0118 - val_m
Epoch 5/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 90ms/step - loss: 0.0109 - mean_absolute_error: 0.0792 - val_loss: 0.0114 - val_m
Epoch 6/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 92ms/step - loss: 0.0111 - mean_absolute_error: 0.0773 - val_loss: 0.0108 - val_m
Epoch 7/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 70ms/step - loss: 0.0108 - mean_absolute_error: 0.0776 - val_loss: 0.0123 - val_m
Epoch 8/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 70ms/step - loss: 0.0103 - mean_absolute_error: 0.0774 - val_loss: 0.0125 - val_m
Epoch 9/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 71ms/step - loss: 0.0110 - mean_absolute_error: 0.0775 - val_loss: 0.0112 - val_m
Epoch 10/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 74ms/step - loss: 0.0105 - mean_absolute_error: 0.0782 - val_loss: 0.0110 - val_m
Epoch 11/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 104ms/step - loss: 0.0099 - mean_absolute_error: 0.0765 - val_loss: 0.0117 - val_
Epoch 12/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 4s 69ms/step - loss: 0.0099 - mean_absolute_error: 0.0762 - val_loss: 0.0123 - val_m
Epoch 13/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 2s 70ms/step - loss: 0.0099 - mean_absolute_error: 0.0763 - val_loss: 0.0123 - val_m
Epoch 14/30
31/31 ━━━━━━━━━━━━━━━━━━━━ 3s 69ms/step - loss: 0.0105 - mean_absolute_error: 0.0781 - val_loss: 0.0123 - val_m
Epoch 14: early stopping
Restoring model weights from the end of the best epoch: 6.
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`.

Validation Metrics for DBO:
RMSE: 0.103687 | MAE: 0.086790 | R² Score: -0.1114
```





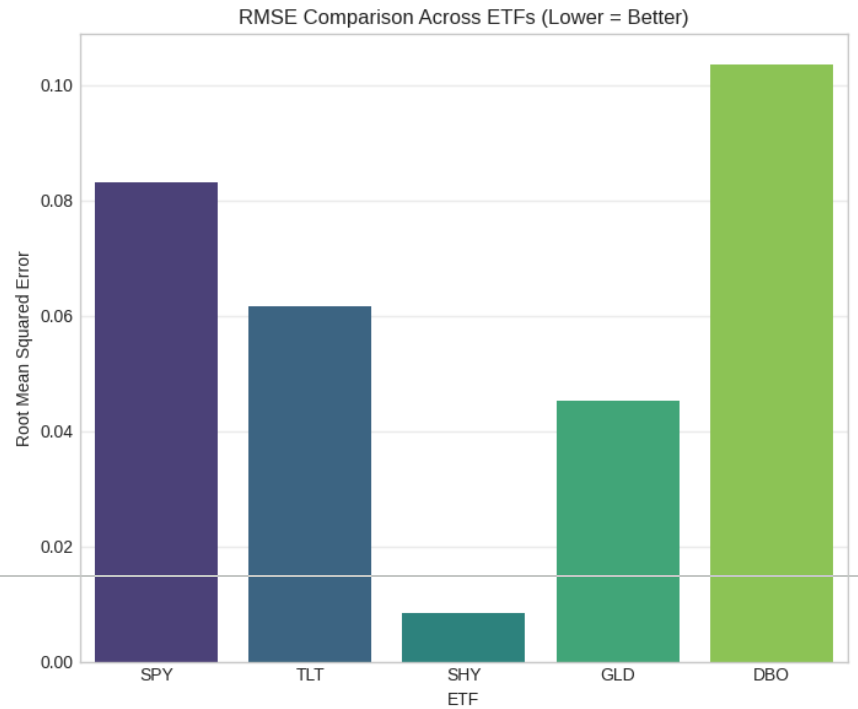

Mean Squared Error
0.018
0.016
0.014
0.012
0.010
0
2
4
6
8
10
12
Epoch

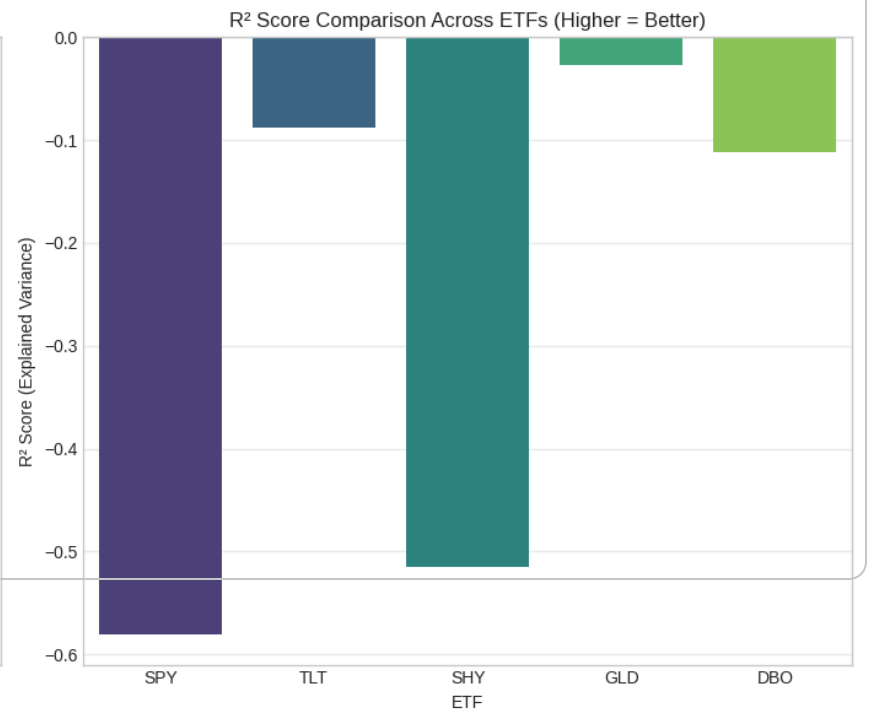

Mean Absolute Error
0.110
0.105
0.100
0.095
0.090
0.085
0.080
0.075
0
2
4
6
8
10
12
Epoch


DBO - Actual vs Predicted 25-Day Ahead Cumulative Return
Actual vs Predicted
Perfect Prediction (y=x)
Predicted 25-Day Cumulative Return
0.2
0.1
0.0
−0.1
−0.2
−0.2
−0.1
0.0
0.1
0.2

Actual 25-Day Cumulative Return

```
=== LSTM Model Performance Summary (All ETFs) ===
   ETF    RMSE     MAE  R² Score
0  SPY  0.0833  0.0720   -0.5809
1  TLT  0.0617  0.0455   -0.0880
2  SHY  0.0085  0.0072   -0.5149
3  GLD  0.0452  0.0367   -0.0273
4  DBO  0.1037  0.0868   -0.1114
```





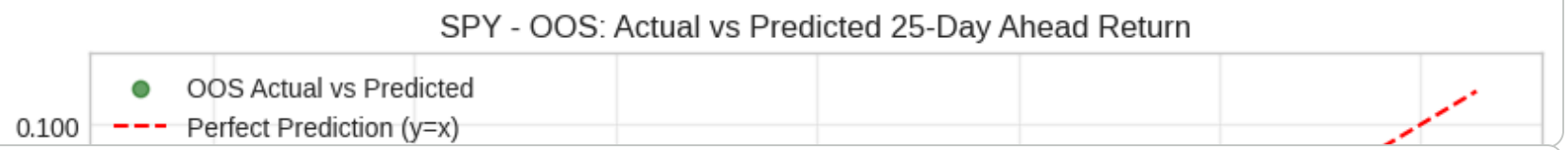
```
def out_of_sample_test(etf, test_start, test_end, lag_window=60, target_horizon=25):
    """
    Conducts out-of-sample testing:
    1. Downloads unseen data (OOS period: e.g., 2023-01-01 to 2023-12-31)
    2. Preprocesses data (returns + target)
    3. Loads trained model and scaler
    4. Generates OOS predictions
    5. Calculates performance metrics

    Parameters:
        etf (str): ETF ticker to test
        test_start (str): OOS period start date (e.g., '2023-01-01')
        test_end (str): OOS period end date (e.g., '2023-12-31')
        lag_window (int): Same lag window used for training
        target_horizon (int): Same target horizon (25 days)

    Returns:
        dict: OOS performance metrics
    """
    print(f"\n=== Out-of-Sample Testing for {etf} ===")

    # Step 1: Download OOS price data (unseen by model)
    oos_price_data = download_etf_data([etf], test_start, test_end)
    if len(oos_price_data) < lag_window + target_horizon:
        raise ValueError(f"Insufficient OOS data: Need at least {lag_window + target_horizon} days")

    # Step 2: Preprocess OOS data (match training pipeline)
    oos_returns = calculate_daily_returns(oos_price_data)
    oos_targets = create_target_variable(oos_returns, target_horizon)

    # Step 3: Create OOS sequences (use same logic as training)
    features = oos_returns[[etf]].rename(columns={etf: 'daily_return'})
    targets = oos_targets[[etf]].rename(columns={etf: 'cumulative_25d_return'})
    aligned_oos_data = features.join(targets, how='inner').dropna()

    X_oos_raw = aligned_oos_data['daily_return'].values
    y_oos = aligned_oos_data['cumulative_25d_return'].values
```

```
# Step 4: Load trained scaler and model (use training scaler to avoid data leakage)
scaler = np.load(f'scaler_{etf}.npy', allow_pickle=True).item()
model = tf.keras.models.load_model(f'lstm_model_{etf}.h5')

# Step 5: Scale OOS features (only transform—DO NOT fit scaler on OOS data!)
X_oos_scaled = scaler.transform(X_oos_raw.reshape(-1, 1))

# Step 6: Create OOS sequences
X_oos = []
for i in range(lag_window - 1, len(X_oos_scaled) - 1):
    sequence = X_oos_scaled[i - lag_window + 1 : i + 1, 0]
    X_oos.append(sequence)
X_oos = np.array(X_oos).reshape(-1, lag_window, 1)
y_oos = y_oos[lag_window:].reshape(-1, 1)

# Step 7: Generate OOS predictions
y_oos_pred = model.predict(X_oos, verbose=0)

# Step 8: Calculate OOS metrics
oos_rmse = np.sqrt(mean_squared_error(y_oos, y_oos_pred))
oos_mae = mean_absolute_error(y_oos, y_oos_pred)
oos_r2 = r2_score(y_oos, y_oos_pred)

print(f"OOS Metrics for {etf}:")
print(f"RMSE: {oos_rmse:.6f} | MAE: {oos_mae:.6f} | R² Score: {oos_r2:.4f}")
print(f"OOS Samples: {len(X_oos)} | Target Range: {y_oos.min():.4f} to {y_oos.max():.4f}")

# Plot OOS actual vs predicted
plt.figure(figsize=(10, 6))
plt.scatter(y_oos, y_oos_pred, alpha=0.6, color='darkgreen', label='OOS Actual vs Predicted')
min_val = min(y_oos.min(), y_oos_pred.min())
max_val = max(y_oos.max(), y_oos_pred.max())
plt.plot([min_val, max_val], [min_val, max_val], 'r--', label='Perfect Prediction (y=x)')
plt.title(f'{etf} - OOS: Actual vs Predicted 25-Day Ahead Return')
plt.xlabel('Actual 25-Day Cumulative Return')
plt.ylabel('Predicted 25-Day Cumulative Return')
plt.legend()
plt.grid(True, alpha=0.3)
plt.savefig(f'oos_prediction_{etf}.png', dpi=300, bbox_inches='tight')
```

```
        plt.show()

        return {
            'ETF': etf,
            'OOS_RMSE': oos_rmse,
            'OOS_MAE': oos_mae,
            'OOS_R²': oos_r2,
            'OOS_Samples': len(X_oos)
        }

    # Update main function to run OOS testing after training
    if __name__ == "__main__":
        # ... (keep previous code for training)

        # Step 5: Out-of-Sample Testing (2023 is unseen data—no overlap with 2018-2022 training/test)
        print("\n=== Starting Out-of-Sample Testing (2023) ===")
        OOS_START = "2023-01-01"
        OOS_END = "2023-12-31"
        oos_metrics = []

        for etf in ETF_SYMBOLS:
            try:
                oos_result = out_of_sample_test(
                    etf=etf,
                    test_start=OOS_START,
                    test_end=OOS_END,
                    lag_window=LAG_WINDOW,
                    target_horizon=TARGET_HORIZON
                )
                oos_metrics.append(oos_result)
            except Exception as e:
                print(f"Error testing {etf} OOS: {str(e)}")
                continue

        # Print OOS Summary
        if oos_metrics:
            oos_df = pd.DataFrame(oos_metrics).round(4)
            print("\n=== OOS Predictive Performance Summary (All ETFs) ===")
            print(oos_df[['ETF', 'OOS_RMSE', 'OOS_MAE', 'OOS_R²', 'OOS_Samples']])
```

```
# Plot OOS vs In-Sample Comparison
in_sample_df = pd.DataFrame(all_metrics)[['ETF', 'RMSE', 'R² Score']].rename(columns={'RMSE': 'InSampl
comparison_df = pd.merge(in_sample_df, oos_df[['ETF', 'OOS_RMSE', 'OOS_R²']], on='ETF')

plt.figure(figsize=(14, 6))
# RMSE Comparison
plt.subplot(1, 2, 1)
comparison_df[['InSample_RMSE', 'OOS_RMSE']].plot(kind='bar', ax=plt.gca())
plt.title('In-Sample vs OOS RMSE (Lower = Better)')
plt.xlabel('ETF')
plt.ylabel('RMSE')
plt.xticks(ticks=range(len(comparison_df)), labels=comparison_df['ETF'], rotation=0)
plt.legend()
plt.grid(axis='y', alpha=0.3)

# R² Comparison
plt.subplot(1, 2, 2)
comparison_df[['InSample_R²', 'OOS_R²']].plot(kind='bar', ax=plt.gca())
plt.title('In-Sample vs OOS R² Score (Higher = Better)')
plt.xlabel('ETF')
plt.ylabel('R² Score')
plt.xticks(ticks=range(len(comparison_df)), labels=comparison_df['ETF'], rotation=0)
plt.legend()
plt.grid(axis='y', alpha=0.3)

plt.tight_layout()
plt.savefig('in_sample_vs_oos_comparison.png', dpi=300, bbox_inches='tight')
plt.show()
```

```
=== Starting Out-of-Sample Testing (2023) ===

=== Out-of-Sample Testing for SPY ===
WARNING:absl:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile_metrics`
Data range: 2023-01-03 to 2023-12-29
Trading days: 250, ETFs: SPY
OOS Metrics for SPY:
RMSE: 0.043249 | MAE: 0.035607 | R² Score: -0.3432
OOS Samples: 194 | Target Range: -0.0571 to 0.1069
```




```
# ----------------------
# Colab Setup (Critical!)
# ----------------------
%matplotlib inline
import os
import warnings
warnings.filterwarnings('ignore')

# Check Colab environment
print("=== Colab Environment Check ===")
print(f"Current Working Directory: {os.getcwd()}")
print(f"Files in Directory: {os.listdir()}")

# ----------------------
# Install Dependencies (if missing)
# ----------------------
try:
    import yfinance as yf
    import pandas as pd
    import numpy as np
    import matplotlib.pyplot as plt
    import seaborn as sns
    from sklearn.preprocessing import StandardScaler
    from sklearn.metrics import mean_squared_error, r2_score, mean_absolute_error
```

```
    import tensorflow as tf
    from tensorflow.keras.models import load_model
    print("All dependencies are already installed.")
except ImportError:
    print("Installing missing dependencies...")
    !pip install yfinance pandas numpy matplotlib seaborn scikit-learn tensorflow --quiet
    import yfinance as yf
    import pandas as pd
    import numpy as np
    import matplotlib.pyplot as plt
    import seaborn as sns
    from sklearn.preprocessing import StandardScaler
    from sklearn.metrics import mean_squared_error, r2_score, mean_absolute_error
    import tensorflow as tf
    from tensorflow.keras.models import load_model
    print("Dependencies installed successfully.")

# Set random seeds for reproducibility
np.random.seed(42)
tf.random.set_seed(42)

# Configure plotting
plt.style.use('seaborn-v0_8-whitegrid')
plt.rcParams['figure.figsize'] = (15, 10)
plt.rcParams['font.size'] = 10
plt.rcParams['axes.grid'] = True

# ----------------------
# Reusable Functions
# ----------------------
def download_etf_data(etf_symbols, start_date, end_date):
    """Download adjusted closing prices with verbose logging"""
    print(f"\nDownloading data for {etf_symbols} ({start_date} to {end_date})...")
    data = yf.download(
        tickers=etf_symbols,
        start=start_date,
        end=end_date,
        interval='1d',
        progress=True,  # Show progress in Colab
```

```
            auto_adjust=False,
            actions=False
        )
        if 'Adj Close' not in data.columns:
            raise ValueError("No 'Adj Close' column found in downloaded data")
        adj_close_prices = data['Adj Close'].dropna()
        print(f"Downloaded: {len(adj_close_prices)} trading days | Columns: {adj_close_prices.columns.tolist()}")
        return adj_close_prices

    def calculate_daily_returns(price_data):
        """Calculate log returns with validation"""
        returns = np.log(price_data / price_data.shift(1)).dropna()
        print(f"Calculated daily returns: {len(returns)} days")
        if len(returns) == 0:
            raise ValueError("Return data is empty (insufficient price data)")
        return returns

    def check_model_files(etf_symbols):
        """Verify pre-trained models and scalers exist (critical for strategy)"""
        print("\n=== Checking for Pre-Trained Model/Scaler Files ===")
        missing_files = []
        for etf in etf_symbols:
            model_path = f'lstm_model_{etf}.h5'
            scaler_path = f'scaler_{etf}.npy'
            if not os.path.exists(model_path):
                missing_files.append(f"Model: {model_path}")
            if not os.path.exists(scaler_path):
                missing_files.append(f"Scaler: {scaler_path}")
        if missing_files:
            raise FileNotFoundError(
                "Missing required model/scaler files (run training code first!):\n" +
                "\n".join(missing_files)
            )
        print("All model/scaler files found ✅")

    # ----------------------
    # Trading Strategy (Colab-Optimized)
    # ----------------------
    def execute_lstm_trading_strategy(oos_start, oos_end, lag_window=60, target_horizon=25):
```

```
etf_symbols = ["SPY", "TLT", "SHY", "GLD", "DBO"]

# Step 1: Validate input files first
check_model_files(etf_symbols)

# Step 2: Download OOS data (verbose)
print("\n=== Starting LSTM Long/Short Strategy ===")
oos_price_data = download_etf_data(etf_symbols, oos_start, oos_end)
oos_daily_returns = calculate_daily_returns(oos_price_data)

# Step 3: Generate rebalance dates (with safety checks)
rebalance_dates = []
all_predictions = []
min_required_days = lag_window + 2 * target_horizon  # Ensure at least 2 rebalance periods
if len(oos_price_data) < min_required_days:
    raise ValueError(
        f"Insufficient OOS data: Need {min_required_days} days, got {len(oos_price_data)}"
    )

# Generate rebalance dates (every 25 days, starting from lag_window)
for i in range(lag_window, len(oos_price_data) - target_horizon, target_horizon):
    rebalance_date = oos_price_data.index[i]
    rebalance_dates.append(rebalance_date)
print(f"Generated {len(rebalance_dates)} rebalance dates: {[d.strftime('%Y-%m-%d') for d in rebalance_date

if not rebalance_dates:
    raise ValueError("No rebalance dates generated (insufficient data range)")

# Step 4: Generate predictions for each rebalance date (verbose)
for idx, rebalance_date in enumerate(rebalance_dates):
    print(f"\n--- Rebalance Date {idx+1}/{len(rebalance_dates)}: {rebalance_date.strftime('%Y-%m-%d')} ---
    period_predictions = {}

    for etf in etf_symbols:
        try:
            # Load model and scaler
            model = load_model(f'lstm_model_{etf}.h5', compile=False)  # Compile=False for Colab compatibi
            scaler = np.load(f'scaler_{etf}.npy', allow_pickle=True).item()
```

```
                # Get data up to rebalance date (no look-ahead bias)
                etf_prices = oos_price_data[etf].loc[:rebalance_date]
                etf_returns = calculate_daily_returns(etf_prices)

                # Prepare input sequence (last lag_window days)
                if len(etf_returns) < lag_window:
                    raise ValueError(f"Only {len(etf_returns)} days of returns (need {lag_window})")
                input_seq = etf_returns.tail(lag_window).values.reshape(-1, 1)
                input_seq_scaled = scaler.transform(input_seq)
                input_seq_3d = input_seq_scaled.reshape(1, lag_window, 1)  # LSTM input shape

                # Predict
                predicted_return = model.predict(input_seq_3d, verbose=0)[0][0]
                period_predictions[etf] = round(predicted_return, 4)
                print(f"  {etf} Predicted 25-Day Return: {predicted_return:.4f}")

            except Exception as e:
                print(f"  ❌ Error predicting {etf}: {str(e)}")
                raise  # Stop execution if any ETF prediction fails

        all_predictions.append(period_predictions)

    # Step 5: Create position signals (verbose)
    print("\n=== Generating Position Signals ===")
    position_signals = pd.DataFrame(
        index=oos_daily_returns.index,
        columns=etf_symbols,
        data=0.0  # Initialize to neutral
    )

    for period_idx, (rebalance_date, predictions) in enumerate(zip(rebalance_dates, all_predictions)):
        # Rank ETFs by predicted return
        ranked_etfs = sorted(predictions.items(), key=lambda x: x[1], reverse=True)
        ranked_names = [etf for etf, pred in ranked_etfs]
        long_etfs = ranked_names[:2]
        short_etfs = ranked_names[-2:]
        neutral_etf = ranked_names[2]

        # Define period end
```

```
        period_end = rebalance_dates[period_idx+1] if period_idx < len(rebalance_dates)-1 else oos_daily_retur
        period_end = min(period_end, oos_daily_returns.index[-1])  # Ensure within bounds

        # Assign weights
        position_signals.loc[rebalance_date:period_end, long_etfs] = 0.5
        position_signals.loc[rebalance_date:period_end, short_etfs] = -0.5
        position_signals.loc[rebalance_date:period_end, neutral_etf] = 0.0

        print(f"Period {period_idx+1}: Long={long_etfs}, Short={short_etfs}, Neutral={neutral_etf}")

    # Step 6: Calculate strategy returns
    print("\n=== Calculating Strategy Performance ===")
    strategy_daily_returns = (position_signals.shift(1) * oos_daily_returns).sum(axis=1)  # Shift to avoid loo
    benchmark_daily_returns = oos_daily_returns.mean(axis=1)

    # Cumulative returns
    strategy_cumulative = (1 + strategy_daily_returns).cumprod() - 1
    benchmark_cumulative = (1 + benchmark_daily_returns).cumprod() - 1

    # Performance metrics
    total_strat_return = strategy_cumulative.iloc[-1] * 100 if not strategy_cumulative.empty else 0
    total_bench_return = benchmark_cumulative.iloc[-1] * 100 if not benchmark_cumulative.empty else 0
    strat_vol = strategy_daily_returns.std() * np.sqrt(252) * 100 if not strategy_daily_returns.empty else 0
    bench_vol = benchmark_daily_returns.std() * np.sqrt(252) * 100 if not benchmark_daily_returns.empty else 0
    strat_sharpe = (strategy_daily_returns.mean() / strategy_daily_returns.std()) * np.sqrt(252) if strategy_d

    # Print performance (clear output)
    print("\n=== FINAL STRATEGY PERFORMANCE (OOS Period) ===")
    print(f"OOS Date Range: {oos_daily_returns.index.min().strftime('%Y-%m-%d')} to {oos_daily_returns.index.m
    print(f"Total Strategy Return: {total_strat_return:.2f}%")
    print(f"Total Benchmark Return (Equal-Weight): {total_bench_return:.2f}%")
    print(f"Annualized Strategy Volatility: {strat_vol:.2f}%")
    print(f"Annualized Benchmark Volatility: {bench_vol:.2f}%")
    print(f"Strategy Sharpe Ratio (Rf=0%): {strat_sharpe:.2f}")

    # Step 7: Plot results (force display in Colab)
    print("\n=== Generating Plots ===")

    # Plot 1: Strategy vs Benchmark Cumulative Returns
```

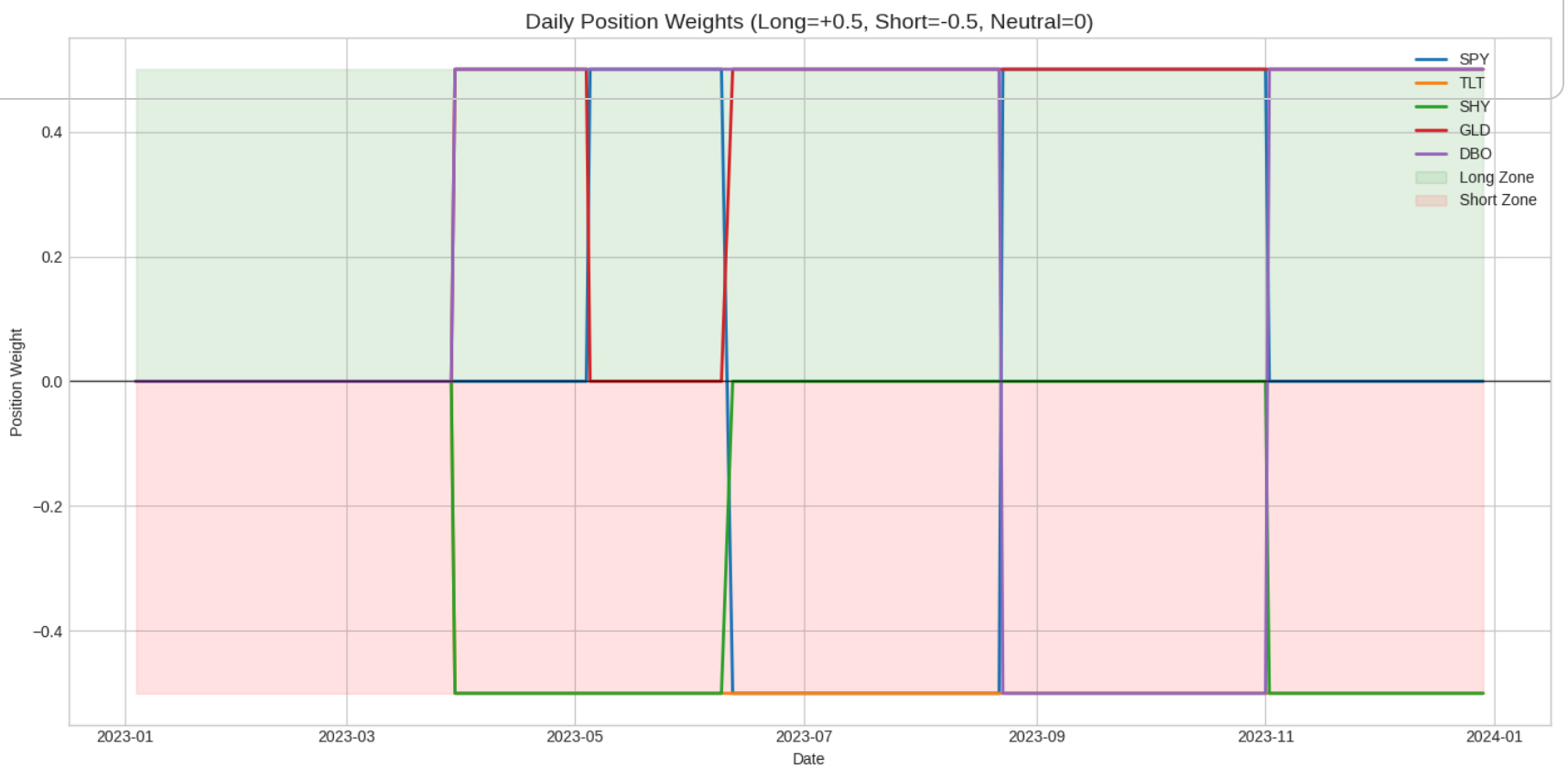
```
    plt.figure(figsize=(12, 6))
    plt.plot(strategy_cumulative.index, strategy_cumulative * 100, label='LSTM Long/Short Strategy', linewidth
    plt.plot(benchmark_cumulative.index, benchmark_cumulative * 100, label='Equal-Weight Benchmark', linewidth
    plt.title('OOS Cumulative Returns: Strategy vs Benchmark', fontsize=14)
    plt.xlabel('Date')
    plt.ylabel('Cumulative Return (%)')
    plt.legend(fontsize=12)
    plt.tight_layout()
    plt.show()  # Force display in Colab

    # Plot 2: Position History
    plt.figure(figsize=(14, 7))
    for etf in etf_symbols:
        plt.plot(position_signals.index, position_signals[etf], label=etf, linewidth=2)
    plt.axhline(y=0, color='black', linestyle='-', linewidth=1, alpha=0.7)
    plt.fill_between(position_signals.index, 0, 0.5, alpha=0.1, color='green', label='Long Zone')
    plt.fill_between(position_signals.index, -0.5, 0, alpha=0.1, color='red', label='Short Zone')
    plt.title('Daily Position Weights (Long=+0.5, Short=-0.5, Neutral=0)', fontsize=14)
    plt.xlabel('Date')
    plt.ylabel('Position Weight')
    plt.legend(fontsize=10, loc='upper right')
    plt.tight_layout()
    plt.show()  # Force display in Colab

    return strategy_daily_returns, position_signals, benchmark_daily_returns

# ----------------------
# Main Execution (Colab-Friendly)
# ----------------------
if __name__ == "__main__":
    # Configuration (match training parameters!)
    OOS_START = "2023-01-01"
    OOS_END = "2023-12-31"
    LAG_WINDOW = 60  # Must match the lag window used during model training
    TARGET_HORIZON = 25  # Must match the target horizon used during model training

    try:
        # Run strategy
        strategy_returns, positions, benchmark_returns = execute_lstm_trading_strategy(
```

```
            oos_start=OOS_START,
            oos_end=OOS_END,
            lag_window=LAG_WINDOW,
            target_horizon=TARGET_HORIZON
        )
        print("\n=== Strategy Execution Completed Successfully! ===")
    except Exception as e:
        # Print full error message (critical for Colab debugging)
        print(f"\n❌ Strategy Execution Failed: {str(e)}")
        import traceback
        traceback.print_exc()  # Show detailed error stack
```

```
=== Colab Environment Check ===
Current Working Directory: /content
Files in Directory: ['.config', 'scaler_DBO.npy', 'strategy_vs_benchmark.png', 'etf_monthly_seasonality.png', '
All dependencies are already installed.
[*********************100%***********************]  5 of 5 completed
=== Checking for Pre-Trained Model/Scaler Files ===
All model/scaler files found ✅

=== Starting LSTM Long/Short Strategy ===

Downloading data for ['SPY', 'TLT', 'SHY', 'GLD', 'DBO'] (2023-01-01 to 2023-12-31)...
Downloaded: 250 trading days | Columns: ['DBO', 'GLD', 'SHY', 'SPY', 'TLT']
Calculated daily returns: 249 days
Generated 7 rebalance dates: ['2023-03-30', '2023-05-05', '2023-06-12', '2023-07-19', '2023-08-23', '2023-09-28

--- Rebalance Date 1/7: 2023-03-30 ---

Calculated daily returns: 60 days
  SPY Predicted 25-Day Return: -0.0013
Calculated daily returns: 60 days
  TLT Predicted 25-Day Return: -0.0376
Calculated daily returns: 60 days
  SHY Predicted 25-Day Return: -0.0041
Calculated daily returns: 60 days
  GLD Predicted 25-Day Return: 0.0077
Calculated daily returns: 60 days
  DBO Predicted 25-Day Return: 0.0177

--- Rebalance Date 2/7: 2023-05-05 ---
Calculated daily returns: 85 days
  SPY Predicted 25-Day Return: 0.0285
Calculated daily returns: 85 days
  TLT Predicted 25-Day Return: -0.0295
Calculated daily returns: 85 days
  SHY Predicted 25-Day Return: -0.0028
Calculated daily returns: 85 days
  GLD Predicted 25-Day Return: 0.0061
Calculated daily returns: 85 days
  DBO Predicted 25-Day Return: 0.0086

--- Rebalance Date 3/7: 2023-06-12 ---
Calculated daily returns: 110 days
```

```
  SPY Predicted 25-Day Return: -0.0044
Calculated daily returns: 110 days
  TLT Predicted 25-Day Return: -0.0329
Calculated daily returns: 110 days
  SHY Predicted 25-Day Return: 0.0005
Calculated daily returns: 110 days
  GLD Predicted 25-Day Return: 0.0037
Calculated daily returns: 110 days
  DBO Predicted 25-Day Return: 0.0355

--- Rebalance Date 4/7: 2023-07-19 ---
Calculated daily returns: 135 days
  SPY Predicted 25-Day Return: -0.0108
Calculated daily returns: 135 days
  TLT Predicted 25-Day Return: -0.0055
Calculated daily returns: 135 days
  SHY Predicted 25-Day Return: 0.0030
Calculated daily returns: 135 days
  GLD Predicted 25-Day Return: 0.0050
Calculated daily returns: 135 days
  DBO Predicted 25-Day Return: 0.0182

--- Rebalance Date 5/7: 2023-08-23 ---
Calculated daily returns: 160 days
  SPY Predicted 25-Day Return: 0.0465
Calculated daily returns: 160 days
  TLT Predicted 25-Day Return: -0.0408
Calculated daily returns: 160 days
  SHY Predicted 25-Day Return: 0.0019
Calculated daily returns: 160 days
  GLD Predicted 25-Day Return: 0.0115
Calculated daily returns: 160 days
  DBO Predicted 25-Day Return: -0.0836

--- Rebalance Date 6/7: 2023-09-28 ---
Calculated daily returns: 185 days
  SPY Predicted 25-Day Return: 0.0512
Calculated daily returns: 185 days
  TLT Predicted 25-Day Return: -0.0243
Calculated daily returns: 185 days
  SHY Predicted 25-Day Return: 0.0026
Calculated daily returns: 185 days
  GLD Predicted 25-Day Return: 0.0067
```

```
Calculated daily returns: 185 days
  DBO Predicted 25-Day Return: -0.0165

--- Rebalance Date 7/7: 2023-11-02 ---
Calculated daily returns: 210 days
  SPY Predicted 25-Day Return: 0.0059
Calculated daily returns: 210 days
  TLT Predicted 25-Day Return: -0.0314
Calculated daily returns: 210 days
  SHY Predicted 25-Day Return: 0.0033
Calculated daily returns: 210 days
  GLD Predicted 25-Day Return: 0.0088
Calculated daily returns: 210 days
  DBO Predicted 25-Day Return: 0.0579

=== Generating Position Signals ===
Period 1: Long=['DBO', 'GLD'], Short=['SHY', 'TLT'], Neutral=SPY
Period 2: Long=['SPY', 'DBO'], Short=['SHY', 'TLT'], Neutral=GLD
Period 3: Long=['DBO', 'GLD'], Short=['SPY', 'TLT'], Neutral=SHY
Period 4: Long=['DBO', 'GLD'], Short=['TLT', 'SPY'], Neutral=SHY
Period 5: Long=['SPY', 'GLD'], Short=['TLT', 'DBO'], Neutral=SHY
Period 6: Long=['SPY', 'GLD'], Short=['DBO', 'TLT'], Neutral=SHY
Period 7: Long=['DBO', 'GLD'], Short=['SHY', 'TLT'], Neutral=SPY

=== Calculating Strategy Performance ===

=== FINAL STRATEGY PERFORMANCE (OOS Period) ===
OOS Date Range: 2023-01-04 to 2023-12-29
Total Strategy Return: -2.92%
Total Benchmark Return (Equal-Weight): 7.46%
Annualized Strategy Volatility: 15.20%
Annualized Benchmark Volatility: 8.91%
Strategy Sharpe Ratio (Rf=0%): -0.12

=== Generating Plots ===
```




```
--- Strategy Execution Completed Successfully! ---
```

Strategy Execution Completed Successfully!

Step 3

```
# Import required libraries
import yfinance as yf
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, r2_score, mean_absolute_error
import tensorflow as tf
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Input, LSTM, Dense, Dropout, BatchNormalization
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
import warnings
warnings.filterwarnings('ignore')

# Set random seeds for reproducibility
np.random.seed(42)
tf.random.set_seed(42)

# Configure plot style
plt.style.use('seaborn-v0_8-whitegrid')
plt.rcParams['figure.figsize'] = (15, 10)
plt.rcParams['font.size'] = 10

# ----------------------
# Data Loading & Preprocessing (Fixed Column Overlap)
# ----------------------
def download_etf_data(etf_symbols, start_date, end_date):
    """
    Downloads daily adjusted closing prices for all 5 ETFs.
    Args:
        etf_symbols (list): List of 5 ETF tickers (SPY, TLT, SHY, GLD, DBO)
        start_date (str): Training start date (YYYY-MM-DD)
        end_date (str): Training end date (YYYY-MM-DD)
```

```
    Returns:
        pd.DataFrame: Adjusted closing prices (index: dates, columns: ETFs)
    """
    data = yf.download(
        tickers=etf_symbols,
        start=start_date,
        end=end_date,
        interval='1d',
        progress=False,
        auto_adjust=False,
        actions=False
    )
    adj_close = data['Adj Close'].dropna()
    print(f"Data Range: {adj_close.index.min().strftime('%Y-%m-%d')} to {adj_close.index.max().strftime('%Y-%m
    print(f"Trading Days: {len(adj_close)}, Assets: {adj_close.columns.tolist()}")
    return adj_close

def calculate_daily_returns(price_data):
    """Calculates daily log returns for all 5 ETFs (handles multi-asset data)."""
    returns = np.log(price_data / price_data.shift(1)).dropna()
    print(f"Daily Returns Shape: {returns.shape} | Columns: {returns.columns.tolist()}")
    return returns

def create_multi_output_targets(returns_data, target_horizon=25):
    """
    Creates 25-day ahead cumulative return targets for ALL 5 ETFs (one target per ETF).
    Adds '_target' suffix to target columns to avoid overlap with feature columns.
    Args:
        returns_data (pd.DataFrame): Daily log returns of 5 ETFs
        target_horizon (int): 25-day prediction window
    Returns:
        pd.DataFrame: Targets (index: dates, columns: <ETF>_target)
    """
    targets = {}
    for etf in returns_data.columns:
        # Cumulative return from t+1 to t+25
        targets[f"{etf}_target"] = returns_data[etf].rolling(window=target_horizon).sum().shift(-target_horizo
    target_df = pd.DataFrame(targets).dropna()
    print(f"Multi-Output Targets Shape: {target_df.shape} | Columns: {target_df.columns.tolist()}")
```